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1.0 Introduction

Company XX is a growing logistics service provider that currently relies fully on manual processes to assign delivery drivers by coordinating through WhatsApp messages to determine driver availability and location. As the order volume starts to increase, this outdated system cannot support the growing demands anymore. The risk of human error increases, and the inefficiencies inherent to such a system severely limits the company's ability to scale and compete in the fast-paced market. Deliveries are inconsistent, customer satisfaction is suffering, and operational transparency is minimal. In an industry where precision and speed are key, the failure to implement a system upgrade would result in the company being outpaced and outperformed, leading to massive potential revenue loss.

To mitigate the growing risks and inefficiencies of the current system, and to support the company's growing demand, we propose a shift to an automated order routing system powered by the latest IT trends – data analytics and machine learning. These innovations will empower the company to meet its rising operational demands with greater performance, increased customer satisfaction, and ultimately securing a stronger competitive advantage for continued growth.

This strategic upgrade will be executed within a 6-month timeline with a capped budget of RM300,000 – RM250,000 as the baseline, and RM50,000 as its management reserves.

2.0 Project Scope Statement (Project charter)

2.1 Project Description

i. In scope activities/ deliverables

1. 3rd-Party API Integration (based on 3rd party API)

The project will integrate Google Maps API by licensing access to its services. This allows our system to leverage Google Maps' existing infrastructure for functionalities and data such as real-time location tracking, navigation, and traffic data, rather than developing a private navigation system from scratch, which would incur greater costs.

2. Integration of Real-Time Driver Location Tracking

The system will leverage the existing driver location data from the external third-party API to display their real-time location. This gives us continuous location updates while negating the need to implement custom GPS hardware into each driver's vehicle.

3. Web-Based Dashboard

A single web-based dashboard will be developed with role-based access, providing tailored interfaces for different user types. In addition, the dashboard has both desktop and mobile views. The latter specifically being for drivers, and the former being for management. The dispatch view (for management) serves to monitor all active drivers and their current statuses (e.g., picked

up, in transit). This provides real-time visibility into the delivery progress of each order together with its performance metrics. The driver view serves to notify, guide, and track individual drivers. It will show the real-time notification for delivery assignments, job details such as pickup location, delivery location, package info, status update – accept, picked up, in transit, delivered, navigation integration (Google Maps).

4. Machine Learning Model for Smart Decision-Making

The machine learning model will take in real-time data such as the drivers' current location, delivery urgency, driver capacity, schedule, preferences, and estimated travel time for each driver, and then determine the most suitable driver for a given order. This ensures maximum efficiency based on objective data.

5. Development of an Automated Driver Assignment System

Core functionality is to assign drivers to orders using machine learning models based on availability, proximity, and other data.

6. User Authentication and Access Control

Since we are developing one web-based application, with different levels of access, we need to have role-based access control. As such, users will sign in using their respective details which signifies their role, then signing in will give them access to their role-based controls accordingly.

7. Training and Documentation

User guidance manuals will be developed and distributed to guide staff and drivers to gain comprehensive understanding of the new system in order to use it to its full potential.

8. System Testing and Pilot Run

Throughout the Software Development Life Cycle, system testing will be conducted rigorously. Towards the end of the deployment stage, a pilot run will be conducted, where the system will be tested in real-life scenarios on a small scale.

9. Deployment and Post-Implementation Support (Within Project Timeline)

Full rollout of the system and short-term support for any issues during the initial phase.

ii. Out of scope activities/ deliverables

1. Developing a Navigation System

It is not within our scope to construct from scratch a navigation system similar to Google Maps for drivers to use and for real-time data gathering.

2. Implementing IoT devices around the country

This goes hand-in-hand with the above activity. Because we are not developing an entire navigation system, we do not need our private IoT devices to feed our private servers real-time information such as driver location, traffic patterns, and etc...

3. Implementing GPS Trackers

Custom GPS trackers attached to each driver's vehicle are out of our scope, as drivers' location will be tracked via the 3rd party API (Google Maps) in the built-in GPS tracker within their phones.

4. Development of Two Separate Dispatch Dashboards Web-Application

Two separate dispatch dashboards – one for management, one for delivery drivers – are redundant. It would cost too much money, effort, and time spent.

5. Redesign of Warehouse or Logistics Workflow

Our system focuses solely on the workflow of the automated order routing system in assigning drivers to orders optimally. As such, it would never touch the physical design of the warehouse or the logistical operations within it.

6. Private Message Driver

Private messages to a customer's order delivery driver is beyond the scope for our current project, although such a feature would be marked as a point of improvement for future updates to the system.

7. Development of a Mobile App

The application is able to be accessible through web-based platforms, but it is not being developed into a dedicated mobile app due to budget and time constraints.

8. Hardware Procurement and Installation

Setting up an infrastructure which includes procurement of servers or tablets to support it is excluded from the scope of our project.

9. Training and Onboarding Programs

Although a user guidance manual will be provided, a full-scale training program will not be provided for all staff as part of this deliverable.

10. Automated Vehicle Dispatching (e.g., using autonomous vehicles)

Only human-driven dispatch is considered in this project. Automation in terms of vehicle hardware or robotics is not part of the deliverables.

iii. Project Objective (SMART Objective)

SMART Criteria	Project Objective
Specific	To design, develop, and implement an Automated Order Routing System (AORS) for a medium-sized logistics company to optimize delivery assignments by utilizing machine learning and real-time data. The system will include a web-based dashboard for dispatchers and drivers, role-based access control, and integration with Google Maps API for real-time routing.
Measurable	The project will: 1) Reduce driver assignment time by at least 70% compared to manual methods 2) Improve on-time delivery rate by 25% within the first month of deployment 3) Achieve ≥85% user adoption (dispatchers and drivers) within 1 month of system rollout 4) Ensure 99% system uptime during operational hours
Achievable	With a budget cap of RM300,000 (RM250,000 baseline + RM50,000 reserves), and the use of external tools (Google Maps API, AWS, Microsoft Project), the goals are achievable within a 6-month Agile development cycle supported by experienced internal team roles (ML engineers, QA testers, DevOps, etc.)
Relevant	This system directly addresses the client's needs to:

	1) Reduce operational workload 2) Improve customer experience with real-time tracking and notifications 3) Support scalability and visibility in order to attract large e-commerce clients like Shopee 4) Reinforce long-term business growth and reduce reliance on manpower
Time-Bound	The full development and testing of the system will be completed within 6 months, from July to December 2025, with the final rollout and user training taking place in Month 6 (December 2025). All SMART outcomes are expected to be evaluated within the first month after deployment

iv. Project Constraints

1. Budget constraint

The total budget allocated for this project is capped at RM300,000, with RM50,000 being management reserves, RM250,000 for development. This naturally limits the scale and complexity of the features within the system.

2. Time Constraint

All the phases are expected to be completed within 6 months, from July to December 2025.

3. Data Availability Constraint

The Machine Learning model heavily relies on past historical data, such as driver performance, to produce accurate predictions on the future. Without the prior existence of these data due to the limitations of the manual system, the Machine Learning model will be inaccurate at its early stages of deployment.

4. Human Resources

Not enough staff and manpower working on the project might cause delays in activity completion, leading to possible delays in project schedule.

5. Stable internet connectivity

A stable internet connection is required to access the web-based dashboard.

v. ***Project Assumptions***

1. Availability of Historical Data

It is assumed that the company has adequate historical data for the machine learning model to be effectively trained on. These data include delivery times, locations, assignments.

2. Stable Internet Connectivity

It is assumed that all drivers always have a stable internet connection on their devices to access the web-based platform.

3. Stakeholder Participation

It is assumed that the key stakeholders — customers, drivers, investors, developers, are all available for regular feedback, testing, and approval at each development stage.

4. Hardware Compatibility

It is assumed that all drivers own a device with hardware capabilities which are compatible to access the web-based system.

vi. ***Critical Success Factor/Measure of Success***

1. Reduction in Delivery Assignment Time

Success is measured by the amount of time decrease in the time taken (eg. 70%) to assign drivers compared to the manual system.

2. Real-Time System Accuracy

Success is also defined by gaining accurate real-time data which are foundational to the project's success in the machine learning model making accurate predictions and smarter choices.

3. User Adoption Rate

At least 85% of all users (drivers and management) fully switch to the system within the first month, it implies that the system is well-constructed, easy to use, useful, and trusted to do the job better than the older method. A high adoption rate also shows that the system has successfully solved a real pain joint issue and that its solution has been effective.

4. Improved On-Time Delivery Rate

Increase in deliveries being delivered on, or before the promised time, as tracked by the system analytics.

5. System Uptime and Reliability

System must be consistently available 99% of the time during operating hours to ensure uninterrupted operations.

2.2 Project Implementation Approach

For the Automated Order Routing System, Agile methodology will be used as the chosen SDLC methodology. The choice is derived from how the on-demand delivery industry has evolved, requiring frequent updates, regular customer feedback, and flexibility to thrive.

The reasons behind selecting Agile methodology for the SDLC of the project are:

1. Iterative and Incremental Delivery
 - a. The development of the system will proceed in short cycles where each iteration will result in a working, testable increment of the product. Such iterations enable early validation, frequent releases, and reduce launch time.
2. Flexible adaptability to changes
 - a. As demand patterns in logistics are unpredictable, Agile makes it possible for the team to modify priorities using insights from customers, real data or recent changes in the market.
3. Stakeholder Engagement
 - a. Regular reviews and demos ensure stakeholders are visible to the project's progress and can influence the current development process.
4. Team Collaboration
 - a. To strengthen engagement between team members open, transparent communication is supported focusing on shared objectives. The Agile framework ensures all technical, functional, and strategic goals are aligned by promoting close teamwork between developers, testers, analysts, and product owners.
5. Risk Mitigation
 - a. External API's integration, cybersecurity compliance and cloud infrastructure scalability are identified and managed incrementally.
6. Value-Driven Prioritization
 - a. Features are prioritized based on business values, urgency and complexity, ensuring demand prediction and route optimization are handled first.

Iteration Structure:

An iteration takes about 2-4 weeks providing enough time for planning, development, testing and reviews to be covered. The fixed length iterations support coherent delivery and monitoring progress over time.

Agile Practices:

The project incorporates key Agile practices to ensure structure and alignment:

- Planning Sessions: At the beginning of every iteration, the team collaborates to define what will be delivered and how it would be built, focusing on the delivery of highly prioritized features first.
- Iteration Reviews: At the end of each iteration the team shares the completed functionality with stakeholder and gather feedback for future improvements and refinements
- Retrospective: Team meeting at the end of each sprint to identify pros and cons, process reflection, and areas of improvement encouraging continuous learning and team growth.

Main Artifacts:

The main artifacts and deliverables conducted from Agile events are:

- Product Backlog: prioritized list for all functions, requirements and features for the project.
- Iteration Backlog: Selecting tasks from the product backlog that must be completed during a specific sprint.
- Product Increments: The shippable product functionality delivered at the end of each iteration, as increments are collected and the sum of increments at the end of the iteration is the final product. It ensures long-term benefits, supports changes, reduces risks, and enhances the final product.

Roles and Responsibilities:

In order to have an organized, well-planned work environment, responsibilities are as follows:

- Development team (not outsourced): Developers, testers and analysts responsible for implementing, testing, integrating the software and delivering the shippable product increments at the end of each iteration.

Due to the complexity in integrating data analytics, cybersecurity and procurement products like Microsoft Project and AWS cloud hosting, Agile remains structured yet flexible. It allows the team to continuously refine the routing algorithm based on real-time data analysis and integrate smoothly with third parties such as Google Maps API and other routing APIs. It helps in collecting feedback from stakeholders for demand forecasting to allocate drivers and warehouses properly, and assists in regulating regular improvements to secure data, information and locations received or sent to customers.

2.3 Work Breakdown Structure

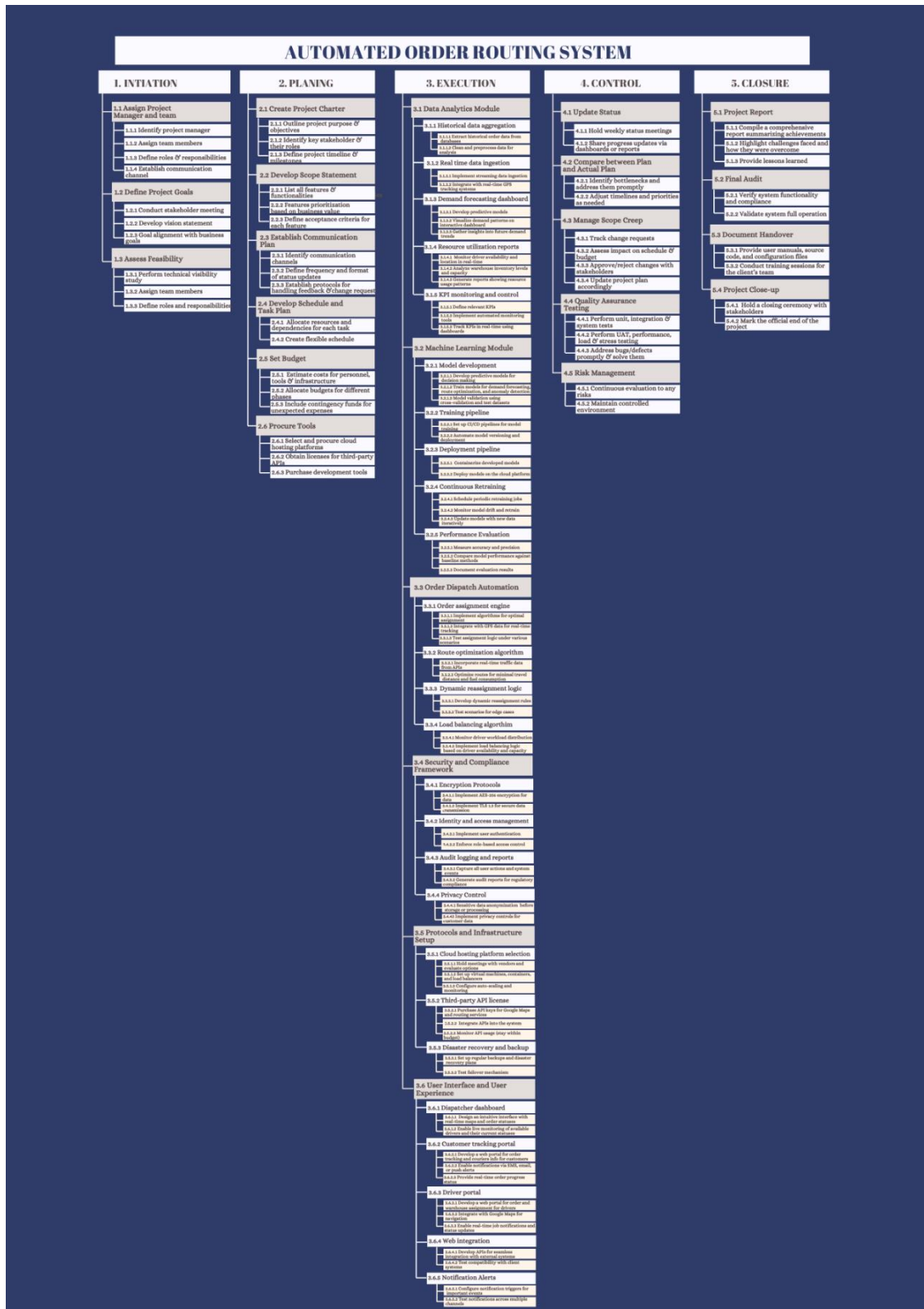


Figure – Work Breakdown Structure

Link to the WBS: https://www.canva.com/design/DAGp65qMaXs/gk6q-JVrGdxdjCES7bgRGQ/edit?utm_content=DAGp65qMaXs&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

1.0 Initiation

1.1 Assign Project Manager and team

1.1.1 Identify project manager

1.1.2 Assign team members

1.1.3 Define roles and responsibilities

1.1.4 Establish communication channel

- **Description:** A skilled project manager needs to be identified and assigned to manage the SDLC of the project. Building a team including developers, data analyst, UI/UX designers, testers and business analysts. Assign specific tasks for each member (Product Owner, Developers, etc.) based on the required skills. Select communication methodology and tools for smooth collaboration among stakeholders. Ending up with a well-structured team formed with each person knowing their role and responsibilities and ready to proceed with the project.

1.2 Define Project Goals

1.2.1 Conduct stakeholder meeting

1.2.2 Develop vision statement

1.2.3 Goal alignment with business goals

- **Description:** Clear identification of project objectives highlighting system main goals such as reducing manual intervention, improving delivery times, enhancing customer experience, and optimizing resource allocation. The goals need to support the client's business purpose and long-term plans focusing on system efficiency, scalability, and user experience. Conduct meetings with stakeholders to ensure clear requirements and function gathernment. Ending up with well-defined goals for the project that will guide future planning and execution.

1.3 Assess Feasibility

1.3.1 Perform technical visibility study

1.3.2 Assign team members

1.3.3 Define roles and responsibilities

- **Description:** Evaluating the chance that the project will be executed as required by the available time, budget and resources. This includes assessing technical feasibility, market readiness, and resource availability considering compliance with regulatory standards. Running a feasibility study will help in checking if the suggested solutions are possible to carry out while analyzing existing tools to identify gaps and areas of enhancement. Ending up with a feasibility report to prove whether the project is feasible enough or points out the areas that need to be changed or improved to reach full vision clarity.

2.0 Planning

2.1 Create Project Charter

2.1.1 Outline project purpose & objectives

2.1.2 Identify key stakeholder & their roles

2.1.3 Define project timeline & milestones

- **Description:** Outline project scope, objectives, stakeholders, high level deliverables and successes factors. The charter will be a reference to stakeholders and team members providing clarity about project scope, goals and expected outcomes. Ending up with a detailed project charter to support the planned activities.

2.2 Develop Scope Statement

2.2.1 List all features & functionalities

2.2.2 Features prioritization based on business value

2.2.3 Define acceptance criteria for each feature

2.2.4 Develop SRS document

- **Description:** Developing a detailed document outlining project boundaries to help manage scope creep. List all features and functionalities to be developed (third-party API integration, real-time driver location tracking, machine learning models, etc.), prioritize features based on business values and complexity, and define the acceptance criteria for each feature. Ending up with a scope statement outlining project deliverables and constraints.

2.3 Establish Communication Plan

2.3.1 Identify communication channels

2.3.2 Define frequency and format of status updates

2.3.3 Establish protocols for handling feedback & change request

- **Description:** Design a plan to determine the best way stakeholders and team members can exchange information and thoughts to ensure transparency throughout the project. Identify communication channels, how frequently the guidelines should be reviewed and how requests for changes are handled.

2.4 Develop Schedule and Plans

2.4.1 Allocate resources and dependencies for each task

2.4.2 Create flexible schedule

2.4.3 Establish Quality Assurance Plan

2.4.4 Create Use case Templates and Testing Plan Framework

2.4.5 Develop Risk Mitigation Plan

- **Description:** Create detailed schedule for the project timeline, defining milestones and assigning deadlines to track project progress effectively and ensure timely delivery. Create use cases templates and testing plan frameworks to ensure quality assurance testing activities are well-planned. Develop risk mitigation plan to assess risk mitigation management in the control phase. Using Agile methodology, iterations will provide flexible timeline and splitting large tasks into small ones helps to handle tasks easily. Well-defined and planned tasks and documentation will help in project management and time constraints.

2.5 Set Budget

2.5.1 Estimate costs for personnel, tools & infrastructure

2.5.2 Allocate budgets for different phases

2.5.3 Include contingency funds for unexpected expenses

- **Description:** Identify all the financial resources the project needs (costs for personnel, tools, infrastructure, third-party services, etc.). Estimate the cost of each component, estimated budget for each phase (initiation, planning, execution, etc.), and include extra budget criteria for emergencies. Ending up with an effective budget that supports future sustainability.

2.6 Procure Tools

2.6.1 Select and procure cloud hosting platforms

2.6.2 Obtain licenses for third-party APIs

2.6.3 Purchase development tools

- **Description:** Select necessary tools and technologies required for project development, testing, and deployment including the allocation of cloud hosting platform, obtaining licenses for third-party APIs (Google Maps, routing services servers), and purchasing any development tools required. Ending up with all the essential tools and technologies present allowing a smooth execution process.

3.0 Execution

3.1 Data Analytics Module

3.1.1 Historical data aggregation

3.1.1.1 Extract historical order data from databases

3.1.1.2 Clean and preprocess data for analysis

- **Description:** Gather and organize historical order data from different sources (databases, logs) to provide access to a clean, organized data set for analysis. Reprocess and clean data so there are no inconsistencies; ending up with a properly arranged dataset that is ready to be analyzed.

3.1.2 Real time data ingestion

3.1.2.1 Implement streaming data ingestion

3.1.2.2 Integrate with real-time GPS tracking systems

- **Description:** Setting up a pipeline that captures real time data from coming orders, selecting the warehouse, then assigning the most suitable courier based on the driver's current location. Ensure low latency and high throughput for real-time processing with high speed, and accurate performance. Ending up with a system that can transfer live data straight into the analytics module in real time.

3.1.3 Demand forecasting dashboard

3.1.3.1 Develop predictive models

3.1.3.2 Visualize demand patterns on interactive dashboard

3.1.3.3 Gather insights into future demand trends

- **Description:** Developing a dashboard that illustrates expected demand trends using both historical and real-time data assisting stakeholders in deciding the best way to use resources and organize operations. Forecasting demand with statistical models will result in a dashboard that offers clear predictions about future needs.

3.1.4 Resource utilization reports

3.1.4.1 Monitor driver availability and location in real-time

3.1.4.2 Analyze warehouse inventory levels and capacity

3.1.4.3 Generate reports showing resource usage patterns

- **Description:** Generate reports that track the driver and warehouse utilization to allocate resources more effectively and efficiently. By monitoring driver availability and current location, checking warehouse inventory and capacity, the generated report will guide resource management.

3.1.5 KPI monitoring and control

3.1.5.1 Define relevant KPIs

3.1.5.2 Implement automated monitoring tools

3.1.5.3 Track KPIs in real-time using dashboard

- **Description:** Develop a mechanism to monitor key performance indicators such as delivery time, accuracy, and cost efficiency ensuring optimal system performance and compliance to defined standards. Implement automated monitoring tools to track KPIs in real-time, ensuring continuous visibility into system performance.

3.2 Machine Learning Module

3.2.1 Model Development

3.2.1.1 Develop predictive models for decision making

3.2.1.2 Train models for demand forecasting, route optimization, and anomaly detection

3.2.1.3 Model validation using cross-validation and test datasets

- **Description:** Develop predictive models using machine learning algorithms to enhance decision-making. The models will analysis the real-time data of driver's current location, delivery urgency, driver capacity, schedule, preferences, and estimated travel time to determine the most applicable courier for a given order. Train models to support leading retailers daily tasks such as predicting demand, route optimization and spotting anomalies. Ending up with ML models providing precise forecast.

3.2.2 Training pipeline

3.2.2.1 Set up CI/CD pipelines for model training

3.2.2.2 Automate model versioning and deployment

- **Description:** Focus on automating the process of ML training and validation to ensure continuous model evolution and performance enhancement over time. Resulting in a training pipeline that allows for ongoing development of the models.

3.2.3 Deployment pipeline

3.2.3.1 Containerize developed models

3.2.3.2 Deploy models on the cloud platform

- **Description:** Use the trained ML models in real-time applications by containerizing models and deploying them on cloud platforms to ensure they are accessible and functional in real-world scenarios. Deployed ML models integrated within the system is applicable.

3.2.4 Continuous Retraining

3.2.4.1 Schedule periodic retraining jobs

3.2.4.2 Monitor model drift and retrain

3.2.4.3 Update models with new data iteratively

- **Description:** Regular updates to the ML models with updated data sets to maintain accuracy, ensuring models adaptability to changes while remaining effective and efficient overtime. Schedule regular retraining and monitor model drifts against changes and time. Resulting in flexible, adaptable models able to respond to environmental changes.

3.2.5 Performance Evaluation

3.2.5.1 Measure accuracy and precision

3.2.5.2 Compare model performance against baseline methods

3.2.5.3 Document evaluation results

- **Description:** Compare ML models performance against predefined metrics to validate the model meets the required accuracy and reliability. By comparing model performance against baseline methods, evaluated evidence of model effectiveness will be established.

3.3 Order Dispatch Automation

3.3.1 Order assignment engine

3.3.1.1 Implement algorithms for optimal assignment

3.3.1.2 Integrate with GPS data for real-time tracking

3.3.1.3 Test assignment logic under various scenarios

- **Description:** Assign orders to drivers by using logic that considers proximity, warehouse item availability and location, capacity, and delivery time. Integrate GPS data for route live tracking and design algorithms for optimal assignment. This reduces manual effort and ensures best order assignment.

3.3.2 Route optimization algorithm

3.3.2.1 Incorporate real-time traffic data from APIs

3.3.2.2 Optimize routes for minimal travel distance and fuel consumption

- **Description:** Optimize routes to reduce travelling distance, time and fuel consumption, enhancing delivery efficiency and minimizing operational cost. Integrate real-time traffic from APIs assisting in route optimization tracks, targeting to improve the delivery routes for greater efficiency.

3.3.3 Dynamic reassignment logic

3.3.3.1 Develop dynamic reassignment rules

3.3.3.2 Test scenarios for edge cases

- **Description:** Reassign orders automatically in the presence of any delay or cancellation, ensuring orders are always assigned to the most suitable courier to

the nearest warehouse, although any distribution occurs. Include different use case scenarios such as sudden traffic jams, last minute order cancelation, delivery delay, etc. Ending up with resilient system that adapts to challenges and distribution.

3.3.4 Load balancing algorithm

3.3.4.1 Monitor driver workload distribution

3.3.4.2 Implement load balancing logic based on driver availability and capacity

- **Description:** Ensure fair and even distribution of workloads among drivers to avoid overloading and to increase drivers' productivity. Implement load balancing logic based on drivers' availability and capacity. Ending up with balanced working environment aiming in maximizing workers productivity.

3.4 Security and Compliance Framework

3.4.1 Encryption Protocols

3.4.1.1 Implement AES-256 encryption for data

3.4.1.2 Implement TLS 1.3 for secure data transmission

- **Description:** Secure sensitive and private data using encryption techniques to protect users' information from unauthorized access.

3.4.2 Identity and access management

3.4.2.1 Implement user authentication

3.4.2.2 Enforce role-based access control

- **Description:** Implement Role-Based Access Control to the system where access is based on roles (driver, customer) and permissions ensuring only authorized personnel can access specific features.

3.4.3 Audit logging and report

3.4.3.1 Capture all user actions and system events

3.4.3.2 Generate audit reports for regulatory compliance

- **Description:** Carry on logs of all system activities for auditing and compliance purposes to verify system transparency and accountability. Generate audit reports for regulatory compliance.

3.4.4 Privacy Control

3.4.4.1 Sensitive data anonymization before storage or processing

3.4.4.2 Implement privacy controls for customer data

- **Description:** Handle personal data by complying with privacy regulations to protect, secure user private data and avoid legal conditions.

3.5 Protocols and Infrastructure Setup

3.5.1 Cloud hosting platform selection

3.5.1.1 Hold meetings with vendors and evaluate options

3.5.1.2 Set up virtual machines, containers, and load balancers

3.5.1.3 Configure auto-scaling and monitoring

- **Description:** Select a cloud provider that encounters scalability, cost, and security requirements. Ensure system stable speed, efficiency and overall performance handling growing demands and high traffic load. Explore AWS and set up virtual machines, containers and load balancers for the infrastructure to accomplish scalable cloud infrastructure that supports system growth.

3.5.2 Third party API license

3.5.2.1 Purchase API keys for Google Maps and routing services

3.5.2.2 Integrate APIs into the system

3.5.2.3 Monitor API usage (stay within budget)

- **Description:** Purchase necessary licenses for external APIs (Google Maps, routing services) so the system can use existing infrastructure instead of developing custom solutions, and track APIs usage to stay within budget. Ending up with reliable third-party servers to enhance system functionality.

3.5.3 Disaster recovery and backup

3.5.3.1 Set up regular backups and disaster recovery plans

3.5.3.2 Test failover mechanism

- **Description:** Implement regular backups and disaster recovery strategies for the system to stay available during failures and have speed recovery. Ending up with a robust system that has quick recovery from outages.

3.6 User Interface and User Experience

3.6.1 Dispatcher dashboard

3.6.1.1 Design an intuitive interface with real-time maps and order statuses

3.6.1.2 Enable live monitoring of available drivers and their current statuses

- **Description:** Develop a dashboard for dispatchers to manage and monitor orders process while allowing live tracking of delivery progress. Intuitive user interface with real-time maps and order status to enhance operational efficiency.

3.6.2 Customer tracking portal

3.6.2.1 Develop a web portal for order tracking and couriers info for customers

3.6.2.2 Enable notifications via SMS, email, or push alerts

- **Description:** Design a web portal for customers to track their order status and check courier information and order progress. Enable notification through SMS, emails and push alerts. The user-friendly interface will help satisfy clients' needs, improve transparency, and build system-customer trust.

3.6.3 Driver portal

3.6.3.1 Develop a web portal for order and warehouse assignment for drivers

3.6.3.2 Integrate with Google Maps for navigation

3.6.3.3 Enable real-time job notifications and status updates

- **Description:** Design a web portal for drivers with a secure login with role-based access control to get their assigned order details, warehouse location, route delivery, and update the order status. Enable notification through SMS, emails and push alerts. For easier and safer driver access, ensure drivers can view, accept or reject orders, and update order status through their mobile devices.

3.6.4 Web integration

3.6.4.1 Develop APIs for seamless integration with external systems

3.6.4.2 Test compatibility with client systems

- **Description:** The system is a web-based system that can be available online for both desktop and mobile devices. Integrate the system with other existing web applications and platforms to ensure seamless interactions with various ecosystems.

3.6.5 Notification alerts

3.6.5.1 Configure notification triggers for important events

3.6.5.2 Test notifications across multiple channels

- **Description:** Verify notifications alerts are sent to stakeholders for important events and changes ensuring everyone is aware of current and new developments. Ensure notifications are sent to customers and drivers for any order updates. Test notification effectiveness across multiple channels.

4.0 Control

4.1 Update Status

4.1.1 Hold weekly status meetings

4.1.2 Share progress updates via dashboards or reports

- **Description:** Keep stakeholders updated with project progress ensuring transparency and building customer trust by holding regular reviews/demos at the end of each iteration and sharing progress updates through dashboards/reports/emails. Resulting in having a strong relationship with stakeholders involved through clear communication.

4.2 Compare between Plan and Actual Plan

4.2.1 Identify bottlenecks and address them promptly

4.2.2 Adjust timelines and priorities as needed

- **Description:** Tracking actual project progress against the plan and timelines to maintain project progress and identify bottlenecks. Burndown charts help in comparing team assignments against the actual work done, supporting Agile method.

By conducting this comparison, the project remains on track as it aligns with the schedule.

4.3 Manage Scope Creep

4.3.1 Track change requests

4.3.2 Assess impact on schedule & budget

4.3.3 Approve/reject changes with stakeholders

4.3.4 Update project plan accordingly

- **Description:** Monitor and control any project changes preventing any impact on system progress and implementation. Manage changes through a formal change control process, evaluate the potential effects of the changes on project schedule, budget, and required resources. Ensure stakeholders approval for any changes or modification, update documents and plans accordingly. Ending up with a controlled project scope that is aligned with initial project goals and objectives while ensuring project speedy delivery within budget and constraints.

4.4 Quality Assurance Testing

4.4.1 Perform unit, integration and system tests

4.4.2 Perform UAT, performance, load and stress testing

4.4.3 Address bugs/defects promptly and solve them

- **Description:** Validate system meets quality standards by passing all rigorous testing. Identify and fix bugs early to confirm product stability and reliability; unit, integration, system, performance, load and stress, UAT testing is performed. Ending up with a high-quality system that delivers reliable outcomes.

4.5 Risk Management

4.5.1 Continuous evaluation to any risks

4.5.2 Maintain controlled environment

- **Description:** Monitor risk on a regular basis to prevent any errors or defects, ensuring project progress is on track and avoiding any delays. Regularly update the risk mitigation plan as needed. Resulting in a controlled environment minimizing risk exposure.

5.0 Closure

5.1 Project Report

5.1.1 Compile a comprehensive report summarizing achievements

5.1.2 Highlight challenges faced and how they were overcome

5.1.3 Provide lessons learned

- **Description:** Documenting the project outcomes and the lessons gained. The document will guide any future work and show how the current approach was successful. Highlighting challenges faced and how they were solved will provide more insights for the project accomplishment.

5.2 Final Audit

5.2.1 Verify system functionality and compliance

5.2.2 Validate system full operation

- **Description:** Conduct a final review to validate that deliverables meet requirements, solving any errors or drawbacks and ensuring correct system functionality. System is available to be deployed and handed over.

5.3 Document Handover

5.3.1 Provide user manuals, source code, and configuration files

5.3.2 Conduct training sessions for the client's team

- **Description:** Passing over all documentation to client including user manuals, source code, configuration files, and other relevant materials. Conduct training for the client's team members. At this stage, the client should be equipped to work with the system on their own.

5.4 Project Close-up

5.4.1 Hold a closing ceremony with stakeholders

5.4.2 Mark the official end of the project

- **Description:** An event to mark the project completion formally and celebrate project success. Once the project is done, there is an opportunity to reflect on the project journey, outcomes, and areas to improve.

The WBS follows Agile principles ensuring optimizing resources utilization and fast delivery within the 6-month timeframe. By breaking down the project into smaller chunks that can be easily managed, ensuring each sprint can be incrementally delivered while staying flexible and adapting to system changes. Starting up with the **initiation phase** focusing on setting up project foundations, goals and team development through 1-2 iterations. Then, the **planning phase** takes place where detailed planning, scope identification, and communication plans are identified within 2-3 iterations. Moving on to the longest phase, **execution phase**, focusing on incremental system development ensuring features implementation and feasibility. The execution phase will take place from 12 – 18 iterations to develop system features and validate its functionality. **Control phase** will be integrated in each phase through reviews and retrospectives to monitor system progress, validate system quality and standards compliance, and to manage risks. Finally, the **closure phase** will take 1-2 iterations to finalize the project, conduct audits, hand over documents, and deliver the system to the client. Following this approach, a robust Automated Order Routing System can be delivered within a 6-month timeframe aligned with customer needs and principles.

2.4 Key Project Milestone

- **Initiation Completion**

This milestone marks the completion of project initiation, ensuring all foundational elements are ready before the detailed planning starts. To reach this point, the project manager and team members are selected, Project charter, scope statement, communication plan, and risk assessment finalized. Obtaining essential tools and having the stakeholders approval for schedule and budget. Achieving this milestone ensures the project starts with clear objectives and goals, defined roles, approved scope and assigned resources ensuring alignment to plan and avoiding any delays.

- **Planning Completion:**

This milestone marks the completion of a detailed planning phase in which the roadmap of the project has been fully defined and is ready for execution. Work breakdown Structure is developed, timelines and task plans are created, resource allocation is finalized, risk management strategy is documented, and features prioritization is completed. By reaching this milestone, the project has a comprehensive plan allowing the team to proceed confidentially into the next phase.

- **Data Analytics Completion**

This milestone marks the completion of collecting, cleaning, analyzing, and organizing all necessary data required for the system. This includes the aggregation of historical and real time

data from various sources, preparing structured datasets for machine learning models, the development of dashboards and reports to visualize KPI's, and support decision making. Successfully reaching this milestone indicates that the project is set on solid ground in terms of analyzed and organized data, which will guide intelligent decision-making throughout the system.

- **AI Completion**

This milestone marks the achievement of developing, training, validation, and integration readiness for all AI models and algorithms. Ensuring machine learning models cover critical functions such as demand forecasting, route optimization and order assignment. At this point, AI components are developed and validated, ready to be used to improve system capabilities through intelligent automation.

- **Core Functionality Completion**

This milestone marks the completion of developing system core functionalities to proceed for system testing and further enhancements. The integration of Google Maps APIs for current location tracking and route optimization, implementation of machine learning models for order, warehouse, route and courier assignment, development of dispatch dashboard accessible with role-based access, real-time orders and driver tracking, and notifications alerts need to be integrated at the time of reaching this milestone.

- **Development Completion**

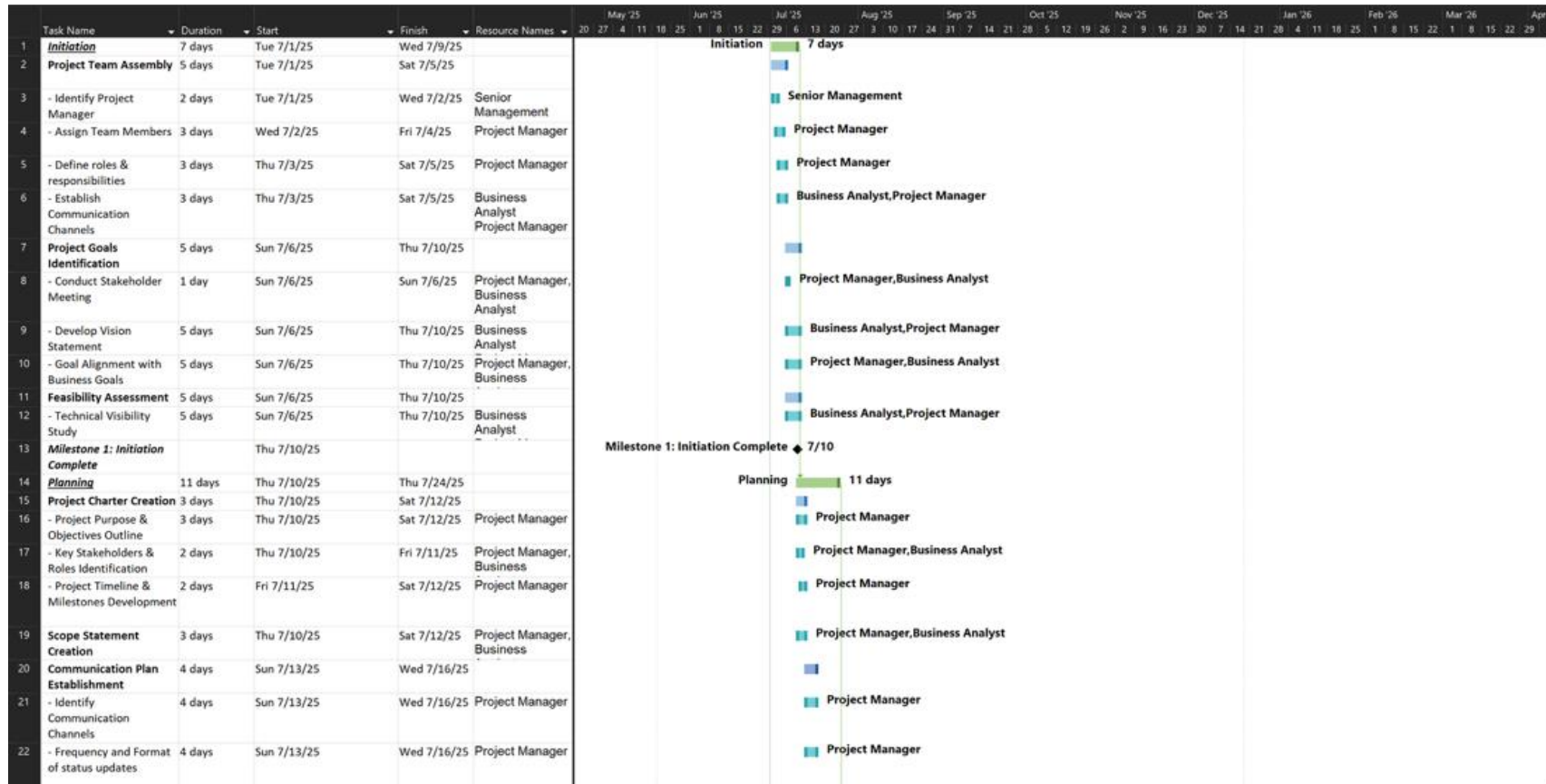
This milestone marks the completion of developing the full system ensuring the implementation of planned features. All other features and integrations are finalized, a complete user authentication and access control is implemented, user documentation and the training materials are ready, final code reviews and quality assurance checks are conducted, system performance and scalability are optimized. By reaching this point, the system is fully developed, ready for rigorous testing and deployment.

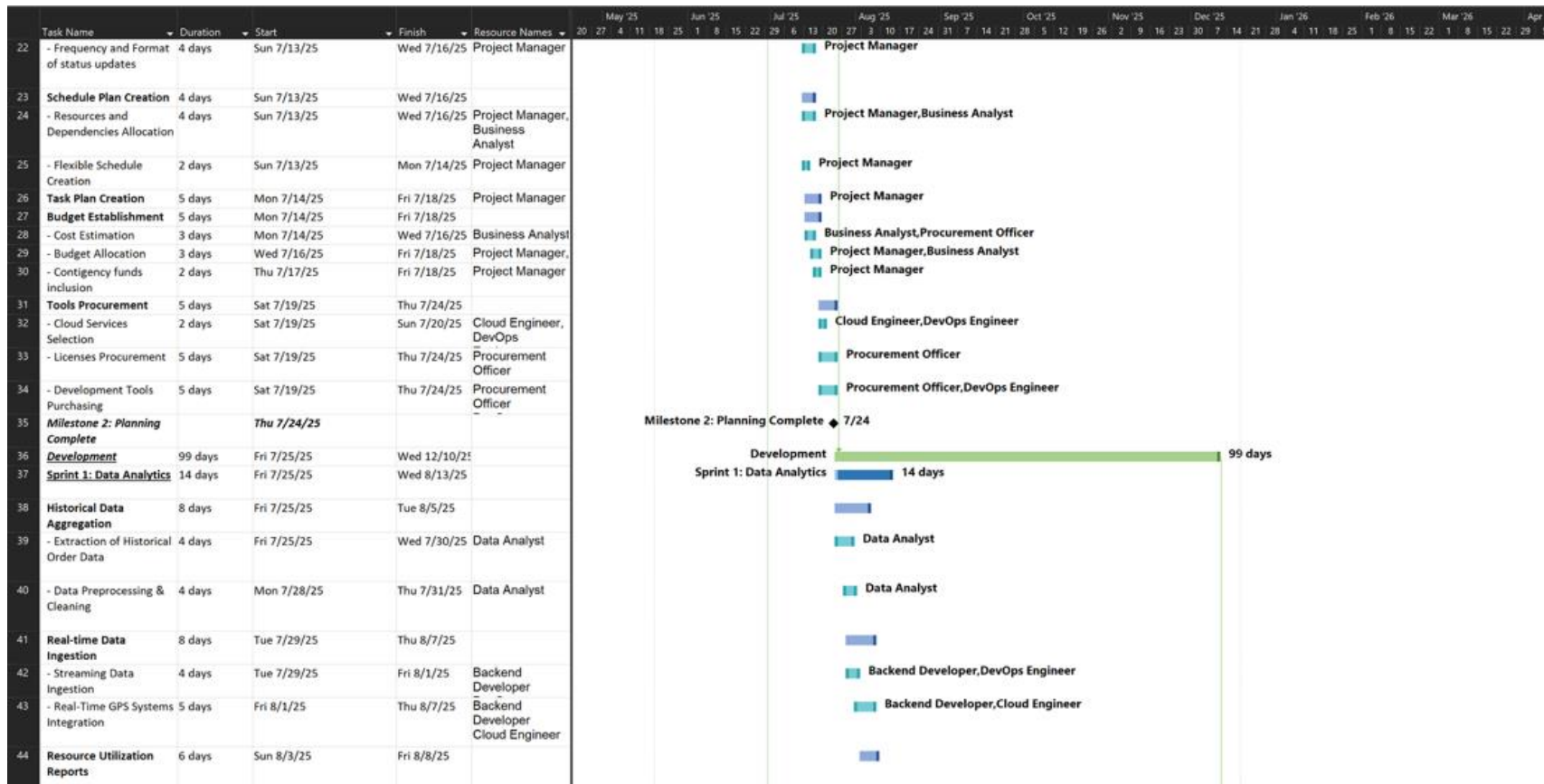
- **Project Completion**

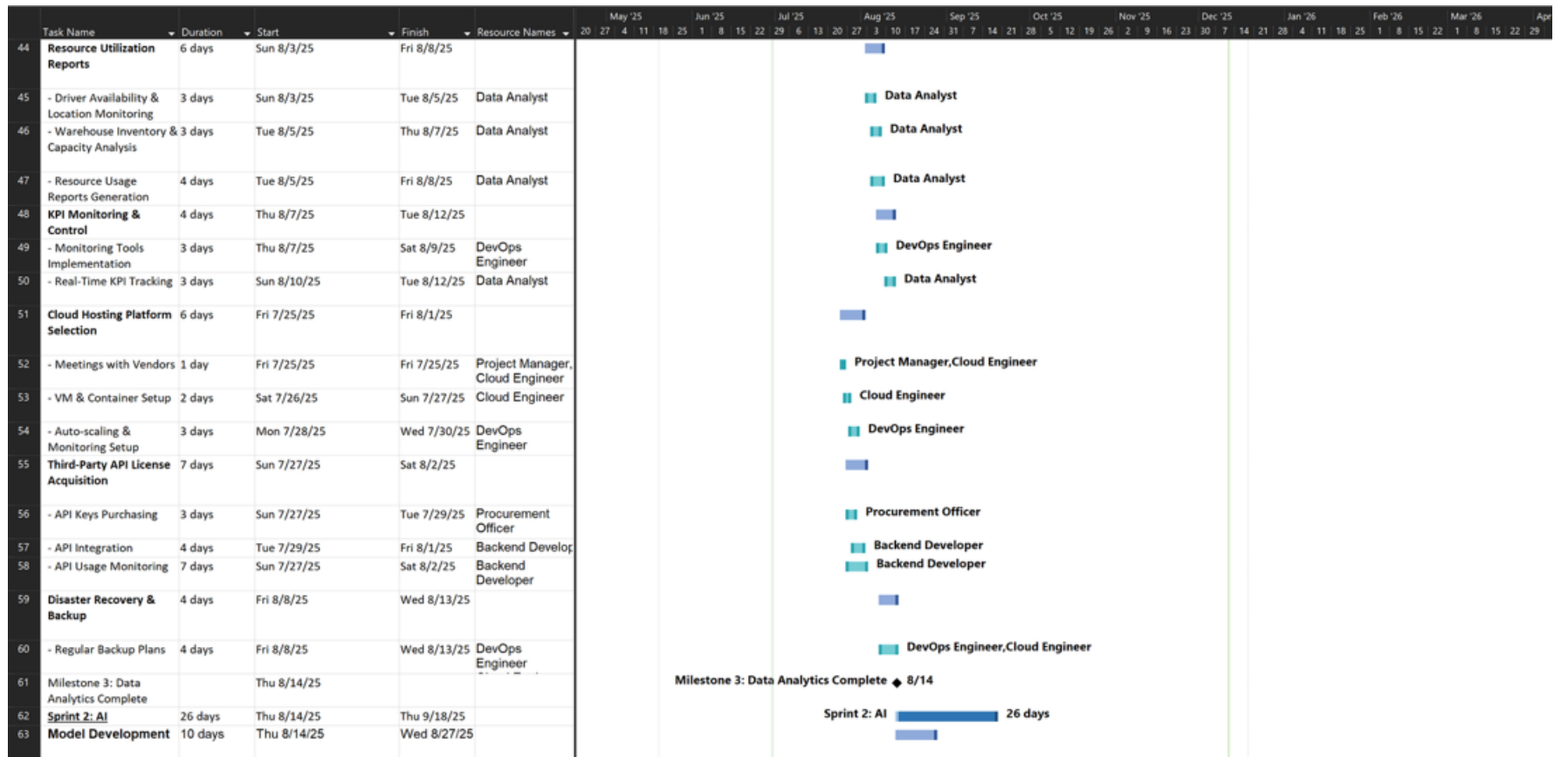
The last milestone to mark the completion and successful delivery of the Automated Order Routing System. The system must be tested under rigorous testing conditions with real-life scenarios to evaluate and validate system performance under various operational conditions before full system deployment. Integration testing, system testing, load and stress testing validating system scalability. Gathering users feedback, fixing bugs, training sessions, and final documentation need to be completed. Stakeholder approval of final product and project close-up meeting marks the official completion of the project.

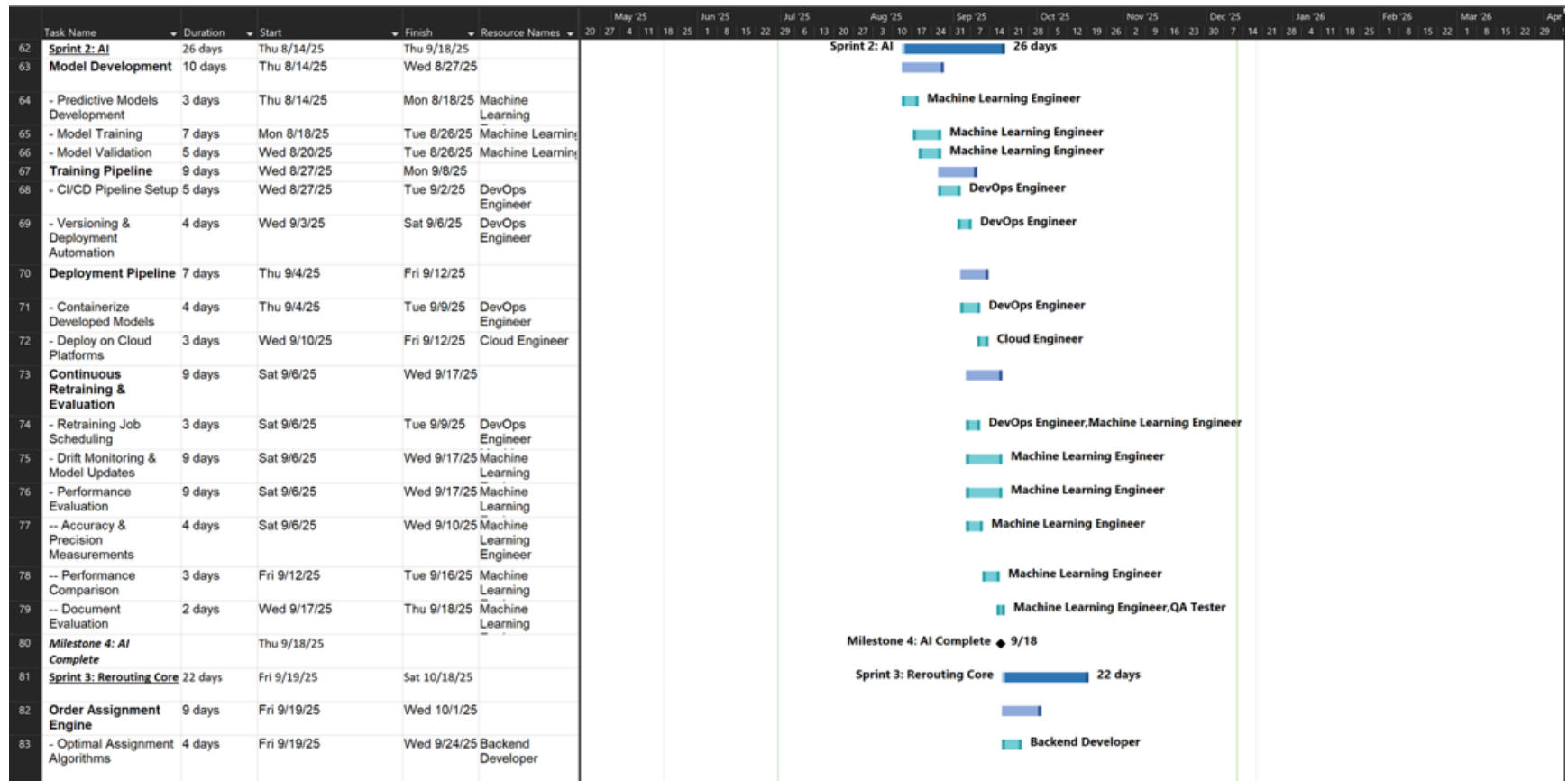
3.0 Project Time Management

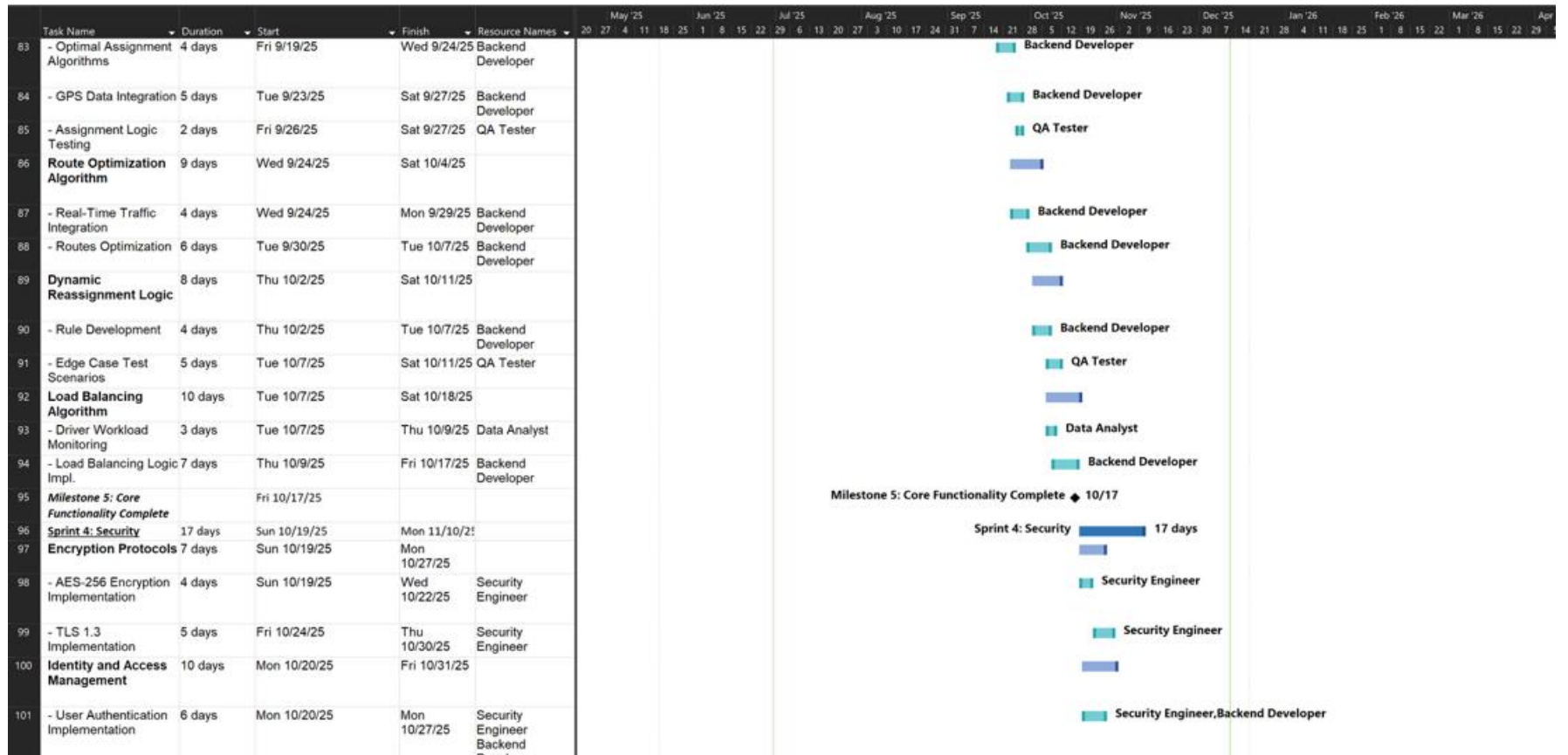
3.1 GANTT Chart

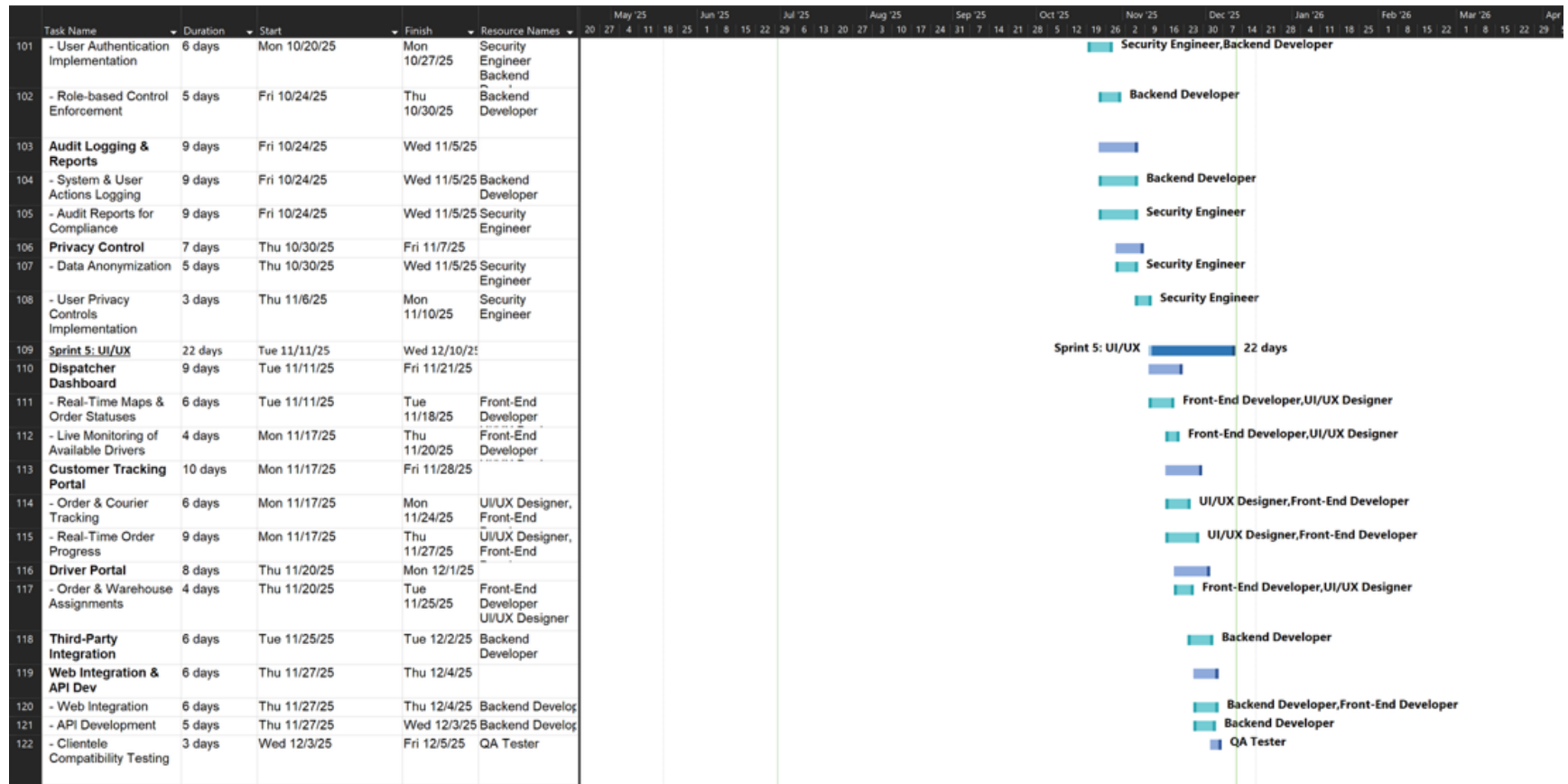


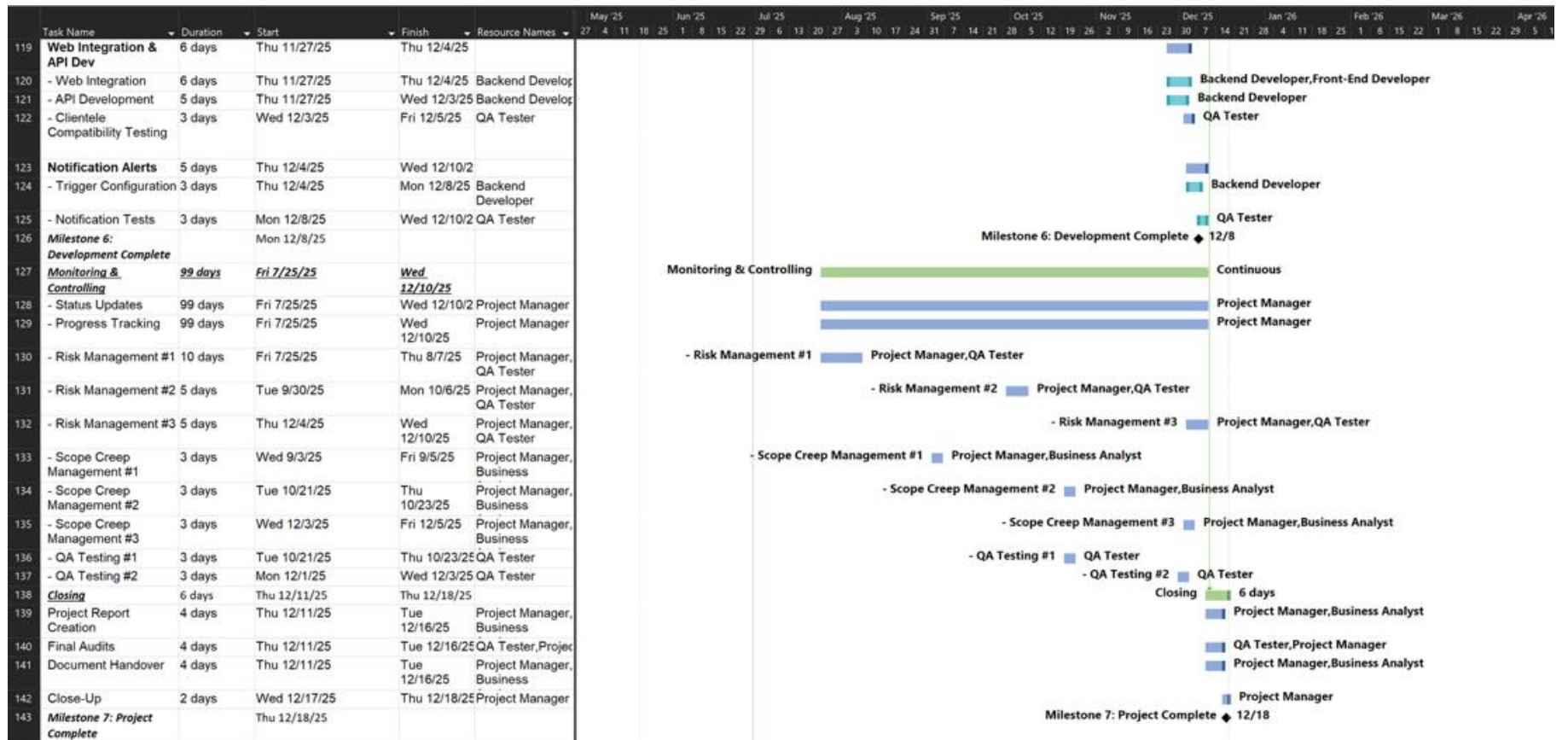












3.2 Schedule Planning Approach

The project schedule was made and modeled directly after the Work Breakdown Structure (WBS) and aligns with the chosen SDLC, Agile. Each sprint has their own distinct functionality and operates independently, unless dependencies are stated. Microsoft Project was used to visualize the plan in a Gantt chart for better clarity.

3.3 Duration Estimation

Analogous Estimation is used to determine and estimate the duration of each activity. The more refined sprints 1 (Data Analytics) & 2 (AI) are used as reference points to aid in the estimation of other activities and sprints such as sprint 3 (Rerouting Core), 4 (Security), 5 (UI/UX) based on complexity, resource availability, and scope. Sprint 1 & 2 themselves have been heavily planned during the project planning phase and their durations have been estimated during that phase.

3.4 Resource Allocation

Task Name	Resource
Initiation	Project Manager, Business Analyst, Senior Management
Project Team Assembly	Project Manager, Senior Management
Project Goals Identification	Project Manager, Business Analyst
Feasibility Assessment	Project Manager, Business Analyst
Planning	Project Manager, Business Analyst, Procurement Officer, DevOps Engineer, Cloud Engineer
Project Charter Creation	Project Manager, Business Analyst
Scope Statement Creation	Project Manager, Business Analyst
Communication Plan Establishment	Project Manager
Schedule Plan Creation	Project Manager, Business Analyst
Task Plan Creation	Project Manager
Budget Establishment	Project Manager, Business Analyst, Procurement Officer
Tools Procurement	Procurement Officer, DevOps Engineer, Cloud Engineer
Development	All Technical Resources
Sprint 1: Data Analytics	Data Analyst, Backend Developer, DevOps Engineer, Cloud Engineer, Procurement Officer
Historical Data Aggregation	Data Analyst
Real-time Data Ingestion	Backend Developer, DevOps Engineer, Cloud Engineer
Resource Utilization Reports	Data Analyst
KPI Monitoring & Control	DevOps Engineer, Data Analyst
Cloud Hosting Platform Selection	Project Manager, Cloud Engineer, DevOps Engineer
Third-Party API License Acquisition	Procurement Officer, Backend Developer
Disaster Recovery & Backup	DevOps Engineer, Cloud Engineer
Sprint 2: AI	Machine Learning Engineer, DevOps Engineer, Cloud Engineer, QA Tester
Model Development	Machine Learning Engineer

Training Pipeline	DevOps Engineer
Deployment Pipeline	DevOps Engineer, Cloud Engineer
Continuous Retraining & Evaluation	Machine Learning Engineer, DevOps Engineer, QA Tester
Sprint 3: Rerouting Core	Backend Developer, QA Tester, Data Analyst
Order Assignment Engine	Backend Developer, QA Tester
Route Optimization Algorithm	Backend Developer
Dynamic Reassignment Logic	Backend Developer, QA Tester
Load Balancing Algorithm	Backend Developer, Data Analyst
Sprint 4: Security	Security Engineer, Backend Developer
Encryption Protocols	Security Engineer
Identity and Access Management	Security Engineer, Backend Developer
Audit Logging & Reports	Security Engineer, Backend Developer
Privacy Control	Security Engineer
Sprint 5: UI/UX	Front-End Developer, UI/UX Designer, Backend Developer, QA Tester
Dispatcher Dashboard	Front-End Developer, UI/UX Designer
Customer Tracking Portal	Front-End Developer, UI/UX Designer
Driver Portal	Front-End Developer, UI/UX Designer
Third-Party Integration	Backend Developer
Web Integration & API Dev	Backend Developer, Front-End Developer, QA Tester
Notification Alerts	Backend Developer, QA Tester
Monitoring & controlling	Project Manager, QA Tester, Business Analyst
Closing	Project Manager, Business Analyst, QA Tester
Project Report Creation	Project Manager, Business Analyst
Final Audits	Project Manager, QA Tester
Document Handover	Project Manager, Business Analyst
Close-Up	Project Manager

3.5 Key Milestones

Milestone	Pre-requisite	Description
1	Initiation Completion	Goal definition, stakeholder alignment, scope identification and high-level requirements gathering completed
2	Planning Completion	Detailed planning finalized, including but not limited to, timelines work breakdown structure, resources and risk management strategy
3	Data Analytics Completion	All necessary data collected, cleaned, analyzed and structured.
4	AI Completion	AI Models and Algorithms developed, trained, validated and ready for integration

5	Core Functionality Completion	Core system features developed and ready for integration
6	Development Completion	Full system development completed and ready for deployment
7	Project Completion	Final testing, documentation and stakeholder approvals completed. Project successfully delivered

4.0 Project Cost Management

4.1 Project Cost Baseline (Cost Estimation)

In this project, the Project Management Team is obtaining average monthly salary information from Jobstreet Malaysia, which began in 1997, one of Asia's top employment data firms with over 10 million job seekers and 90,000 companies, since 1999 it has remained profitable. Despite the presence of competition from LinkedIn and Monster Inc, Jobstreet is highly popular due to ease and significant dominance in Malaysia (Samah et al., 2022). As such, Jobstreet Malaysia was chosen to provide average salary information utilised in this project.

According to the Laws of Malaysia based on Jabatan Tenaga Kerja Semenanjung Malaysia (JTKSM), under the Employment Act 1955, an employee can only work 8 hours a day, 45 hours a week. Hence, in this project, the team utilize 45 hours a week to calculate the monthly wage of each human resource needed in this project.

The average monthly salary of the manpower, as seen below, is taken from JobStreet Malaysia, last updated on 1 June 2025.

According to the official Microsoft website, Project Plan 3 has a cost of \$30 per month, billed annually, approximately RM 127.57 per user per month based on the current exchange rate.

According to the official Google Maps Platform pricing website, until 1 March 2025, the Dynamic Maps API, which categorized under the Essentials plan, has an initial 10,000 complimentary loads per month per SKU (Google, 2025). The API is supportive of interactive maps for user engagement, such as panning, zooming, and markers, which renders it suitable as the tool in this project with real-time task assignments and auto-routing. After the first 10,000 free loads, Google charges based on the number of map loads requested.

$$35,000 \text{ Billable loads} = \$30 \approx \text{RM } 744.19$$

$$\text{Billable loads} = 35,000 - 10,000 = 25,000$$

$$\text{Cost per request} = \frac{\text{RM } 744.19}{25,000} \approx \text{RM } 0.02977 \text{ per map load}$$

During the initial development period, it would be a reasonable assumption that every developer would be producing about 100 map loads a day through debugging, testing, and feature merging. For 3 developers working 22 days a month:

$$3 \text{ devs} \times 100 \text{ requests per day} \times 22 \text{ days} = 6,600 \text{ map loads/month}$$

No cost shall be incurred in 4 months of development if monthly usage is maintained at 6,600 map loads, since $6,600 < 10,000$, which remains in the map loads that are provided in the Essentials plan free tier.

According to the official AWS EC2 On-Demand pricing webpage, the rate for keeping a t3.medium instance, with 2 vCPUs and 4 GiB of RAM in the Asia Pacific (Malaysia) region, costs USD \$0.0659 per hour, which is approximately RM 0.28 per hour using the current exchange rate (AWS, 2025).

$$24 \text{ hours per day} \times 30 \text{ days} = 720 \text{ hours/month}$$

$$RM 0.28 \times 720 \text{ hours} = RM 201.60$$

Item	Cost per month (RM)	No. of months needed	Month	Subtotal (RM)
MANPOWER				
Project Manager	8500.00	6	July-Dec	51,000.00
Software Developer (Back-end)	5800.00	6	July-Dec	34,800.00
QA Tester	4800.00	6	July-Dec	28,800.00
Cloud Engineer	8500.00	3	July-Sep	25,500.00
DevOps Engineer	7500.00	3	July-Sep	22,500.00
Business Analyst	5500.00	4	July, Sep, Oct, Dec	22,000.00
Data Analyst	4800.00	3	July, Aug, Oct	14,400.00
Machine Learning Engineer	5917.00	2	Aug, Sep	11,834.00
Security Engineer	5644.00	2	Oct-Nov	11,288.00
Senior Management	8000.00	1	July	8,000.00
Software Developer (Front-End)	6250.00	1	Nov	6,250.00
UI/UX Designer	5800.00	1	Nov	5,800.00
Procurement Officer	3200.00	1	July	3,200.00
Software				
Microsoft Project (Planner and Project Plan 3)	127.57	1	July	127.57
Google Maps API (Distance Matrix)	0.0	4	July-Dec	0.0
Infrastructure				
AWS EC2 (t3.medium)	201.60	6	July-Dec	1209.60
Project Estimate				246709.17
Contingency Reserve				
Scope Creep				2,000.00
Google Maps API Overuse				500.00
AWS Hosting Overages: Downtime Mitigation				500.00
Project Cost Baseline				249709.17

4.2 Project Cost Budget

Management reserve, unlike contingency reserve, addresses risks that are yet unknown, or "unknown unknowns." Management reserves range from 5 to 15 percent and vary according to the complexity of the project (Prieto, 2021).

The determination of an appropriate level of management reserve is policy-driven within the organization and, more significantly, by the project's overall uncertainty. Whether or not the project is a first of a kind, working in a new country, or the utilization of unfamiliar technology are some of the determinants that must be taken into consideration. Conversely, repeat projects performed for the same or similar clients in the same or similar environments may warrant fewer reserves (Prieto, 2021).

With projects containing AI/ML components (with model performance risks), cloud infrastructure (with hosting and API usage risks), security-sensitive data (with compliance risks), and agile delivery approaches (with requirements change risk), these factors imply a medium to high level of both technical and stakeholder uncertainty. As such, it is prudent to set aside approximately 10 percent of the project budget to ensure the necessary flexibility to address these risks.

Management Reserve: 10%

Project Cost Budget: Cost Baseline + Management Reserve

$$RM\ 249709.17 + 10\ \% = RM\ 274680.10$$

Project Cost Budget: RM 274680.10

5.0 Project Quality Management Plan

The project Quality Management Plan is to ensure that the system meets its performance, accuracy, and usability targets, as outlined in the project's critical success factors. The system involves real-time data processing, machine learning -based decision-making, and external APU integration (such as, Google MAP API, routing service), maintaining consistent quality throughout development and deployment is important.

The objectives of the system are:

- Ensure driver assignment decision are made within 1.5 seconds
- Maintain system uptime of 99% or higher
- Achieve at least 85% adoption rate among users (driver & dispatcher) in the first month
- Deliver accurate dispatch assignment using a trained machine learning model

These objectives will be measured and monitored using two key tools suited for this project: Control Charts and Pareto Charts.

5.1 Quality Tool/Technique

Tool_1: Control Chart

Control charts are statistical tools that monitor the performance of the project's process over time. It shows if the process is stable (under control) or if there are variations that need investigation. It will be used to track and monitor system response time, especially for the automated driver assignment process. By plotting the dispatch latency overtime, the team can observe whether the system remains within acceptable performance thresholds.

Technique Application:

During each sprint, the development team will collect response time logs from the backend when an order is submitted and a driver is assigned, the collected data point will be plotted on a control chart, observing the consistency of the performance over time. Based on the project's performance goal with the target average of 1.5 seconds, the upper control limit (UCL) might be set to 2 seconds. If any data point exceeded the UCL or showed a trend of increasing delays (exp, 5 consecutive slow assignment), it will flag this issue during the Sprint Retrospective.

The control charts will track system response time (goal is to less than 1.5 seconds) and uptime percentage during daily operations.

This tool supports the goal of maintaining ≤ 1.5 seconds dispatch delay and ensures stability in the real-time assignment engine. Besides that, it supports proactive performance tuning during iterative development

Tool_2: Pareto

The Pareto is used to prioritize user-reported bugs and issues during the system testing and pilot deployment phases. It helps the team identify which the problem has the most significant impact on user and prioritize which should be addressed at the first place.

Technique Application:

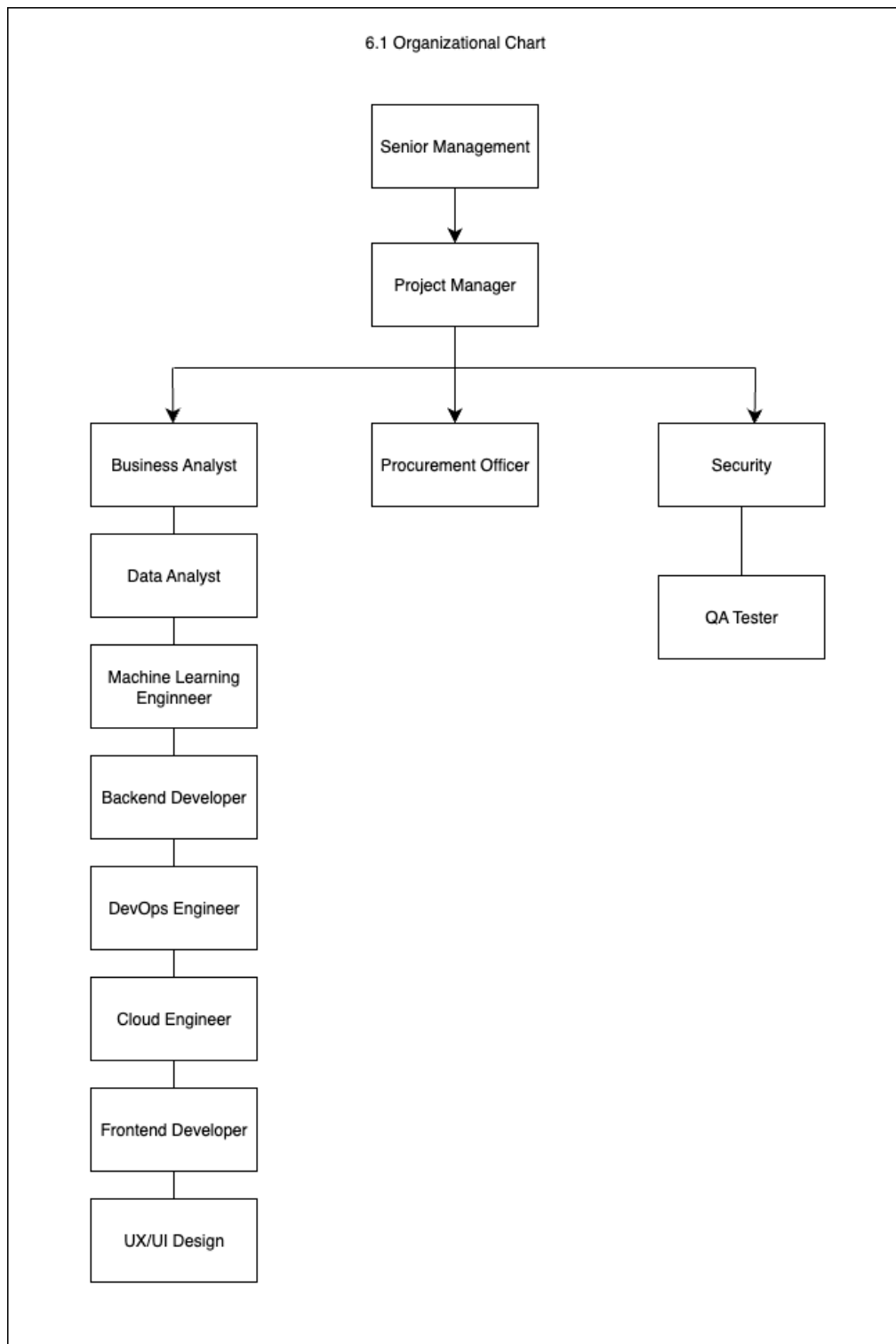
During the pilot testing, users (company management and drivers) will submit feedback regarding errors such as delayed or incorrect driver assignments, dashboard glitches & slow loading, and login authentication problems.

The team will categorize and count these issues, then the implementation of Pareto chart helps with visualizing the most frequent problem category. For example, if more than 70% of user complaints relate to dashboard response times, that issue becomes the top priority in the next sprint backlog.

This technique helps the team focus its limited time and resources on fixing the most impactful issues first, which helps improve user satisfaction and supports the goal of achieving 85% user adoption within the first month.

6.0 Project Human Resources Management Plan

6.1 Project Organization Chart



6.2 Roles and responsibilities

There are specific roles based on the Agile Scrum methodology that implanted in the project, has been assigned to each team members to support the successful development of the project.

Senior Management

This role provided strategic direction and reviewed key milestones throughout the AORS development cycle. The role is responsible for approving the initial scope, confirming alignment with organizational goals, and validating whether the proposed features like dispatch automation and real-time monitoring. It also signed off on the system during final review to confirm business readiness.

Project Manager

Project Manager oversaw all phases of the whole project, from the begin planning to final testing. This role involved task scheduling, sprint coordination, and progress tracking for the development of core modules such as the automated dispatch engine, tracking interface, and dashboard components. This role ensures that each team met the deadline and quality target while facilitating communication between technical roles and stakeholders.

Business Analyst

This role helps on converting the user requirements into technical tasks. Specific deliverable included outlining the logic for driver assignment prioritization, filtering function in the management dashboard (for exp: delivery zones, time slot), and the functional flow of driver job updates. These were documented and passed to the development team for the implementation.

Procurement Officer

This role helps on the acquisition of critical tools and services, it includes the Google Maps API (function for re-routing and geolocation), cloud hosting services, and the productivity tiil for the internal project tracking. The officer helps preventing integration a delay during sprints to ensure these were available ahead of development cycle.

Data analyst

This role helps on processed and structured dataset that required to train and validate for the ML (machine learning) model. These included delivery times, driver availability record, and the zone-based demand trends. This role also supported the creation of dashboard graphs for the management report and collaborated closely with the Machine Learning Engineer to identify the pattern for the improved model accuracy in the future.

Machines Learning Engineer

This role is handled on design and train the predictive model responsible for selecting the most suitable driver based on the entities (live availability, location and past performance). The key task for this role included model training, evaluation using validation data, and exporting prediction into the backed assignment logic. ML Engineer contributed directly to meeting the system's goal of <1.5s dispatch time.

Backend Developer

This role implemented the central system logic, which included the system dispatch algorithms, API endpoints, and the order tracking flows. For example, the integrating of the ML model's output into the dispatch logic, connecting to Google Maps for real-time routing (drivers), and enabling live job update in the system databased in short time.

DevOps Engineer

This role is to built and maintained the CI/CD pipeline that automated deployment of the AORS system update. This included the backend logic for smart driver assignmenr an dthe frontend dashboard UI. They have to ensure the smooth integration of each sprint release into the cloud environment and monitored stability after new builds were pushed life

Cloud Engineer

This role helps on configuring and maintaining the hosting environment for the whole AORS system. The key tasks for the role including setting up the virtual machines, loading balancing , auto-scaling policies, and real-time server monitoring. These ensured the system met uptime goals during the testing phase and could scale during the peak usage period

Fronted Developer

This role built the web-based interface that used by the users (dispatchers and drivers). The responsibilities included developing the real-time dashboard that able to display the job statues, assign driver, and the active route. Besides that, the role handled maintaining the responsive view and interactive component that connected to backend APIs for the live data updates.

UI/UX Designer

This role is worked on the user journey mapping and the UI (interface) layout for user's view (dispatcher and driver). The tasks included designing wireframes, optimizing the layout for the mobiles use(r), and ensuring the visual consistency across all modules. This role is essential and directly improved system usability and supported faster onboarding during user testing phase.

Security Engineer

This role implemented the system-wide security control, which included AES-256 encryption for delivery data and TLS 1.3 for secure API communication. Besides that, they help on configuring role-based access control so that each of the users (dispatchers, admin, and drivers) had proper dashboard access without the data is overlapped, aligning with privacy standards.

QA (Quality Assurance) Tester

This role helps on validating the system's dispatch flow, real-time driver assignment speed, and the dashboard functionality across all roles. They created and executed the test cases for the job matching accuracy, tested fallback scenarios for the driver unavailability, and to ensure the system's stability before the system release to the market

7.0 Project Communication Plan

7.1 Communication Plan

Communication Type	Objective	Medium	Frequency	Audience/Recipient	Owner	Deliverable	Format
Sprint Review Meetings	To show progress and gather feedback from stakeholders	Video Call (Zoom)	End of every sprint (bi-weekly)	Product Owner, Client Stakeholders	Business Analyst	Sprint Review Report	Slide deck, PDF
Internal Daily Stand-ups	To track daily progress and unblock team members	Microsoft Team Chat	Daily (10mins)	Development Team, Scrum Master	DevOps Engineer	Verbal updates	Spoken (Agile Style)
Weekly Status Reports	To update progress and highlight issues/risks.	Email with attachment	Weekly	Project Sponsor, Management, Product Owner	Project Manager	Project Status Report	Document (.docx/.pdf)
Issue Escalation Reports	To report critical issues beyond team resolution	Email, WhatsApp	As needed	Project Manager, Senior Management	QA Tester	Issue Log, Risk Notification	Structured template
Change Request Discussion	To propose or approve changes in scope or features	Zoom + Email Follow-up	Bi-weekly or As needed	Stakeholders, Business Analyst, PM	Business Analyst	Change Request Form	Word Doc + Meeting Notes

8.0 Project Risk Management Plan

Risk is the possibility of loss or injury, and there are several ways to identify project risks, such as brainstorming, interviews, the Delphi method, and SWOT analysis. Our group chose SWOT analysis because it effectively examines both internal and external factors that may impact a project. Strengths show areas of stability but can also pose risks if the project relies too heavily on them. Weaknesses highlight internal risks like skill gaps or limited resources. Opportunities, while beneficial, can become risks if pursued without proper planning. Threats include external factors like market changes or competition. By analyzing each SWOT quadrant, we can identify and manage potential risks early.

8.1 SWOT Analysis

8.1.1 Scope Creep

STRENGTHS	WEAKNESSES
<i>Improved product outcome</i> <i>Enhanced customer satisfaction</i> <i>Responsive to client needs</i>	<i>Budget Overrun</i> <i>Project timeline setbacks</i> <i>Resource Drain</i>
OPPORTUNITIES	THREATS
<i>Team skill growth</i> <i>Scoping Tools Development</i> <i>Change control Process Improvement</i>	<i>Project Failure</i> <i>Team Burnout</i> <i>Client Dissatisfaction</i>

8.1.2 Google maps API overuse

STRENGTHS	WEAKNESSES
<i>Global Coverage and Data Quality</i> <i>Ease of Integration and Developer Tools</i> <i>Scalability</i>	<i>High Cost for advanced features</i> <i>Vendor lock-in</i> <i>Limited flexibility in customization</i>
OPPORTUNITIES	THREATS
<i>Alternative mapping providers</i> <i>Usage Monitoring Tools</i>	<i>Unexpected cost spikes</i> <i>Service Disruptions</i> <i>Security vulnerabilities</i> <i>Changes in Google's API Policy</i>

8.1.3 AWS Hosting Overages: Downtime Mitigation

STRENGTHS	WEAKNESSES
<i>Scalable Infrastructure</i> <i>Monitoring Tools to track performance and usage</i>	<i>High cost</i> <i>Budget Overrun</i> <i>Technical Dependency</i> <i>Over-Reliance on Automation</i>
OPPORTUNITIES	THREATS
<i>Team skill growth</i> <i>Serverless Migration</i>	<i>Security Incidents/API Key Compromise</i> <i>Billing Surprises</i>

8.2 Risk Register

Number	R1
Rank	1
Risk	Scope Creep
Description	Uncontrolled expansion of project scope
Category	Project Scope
Root Cause	Client adds non-critical features mid-project
Triggers	Frequent new requests during development
Potential Responses	Enforce strict change control procedures; track change requests formally
Risk Owners	Project Manager
Probability	Very High
Impact	High
Status	Meeting is scheduled with the stakeholders

Number	R2
Rank	2
Risk	Google Maps API Overuse
Description	Failure to integrate with third-party platforms (e.g., Shopee API)
Category	Technical
Root Cause	Unstable or undocumented external APIs
Triggers	Errors during integration testing
Potential Responses	Build buffer time for integration; coordinate

	early with API providers
Risk Owners	Dev Lead / PM
Probability	High
Impact	Moderate
Status	Early coordination initiated with API vendors; mock testing in progress

Number	R3
Rank	2
Risk	AWS Hosting Overages: Downtime Mitigation
Description	Service interruptions affect routing system performance
Category	Technical
Root Cause	Poor hosting configuration or lack of redundancy
Triggers	User complaints or order backlog spikes
Potential Responses	Use high-availability cloud services; implement monitoring and alerts
Risk Owners	DevOps Engineer
Probability	High
Impact	Moderate
Status	Auto-scaling and alert configurations under review

Number	R4
Rank	4
Risk	Team Skill Growth
Description	Opportunity for team to acquire new technical and management skills during project execution
Category	Human Resource
Root Cause	Exposure to new technologies and project challenges
Triggers	Assignment of new responsibilities, training involvement
Potential Responses	Exploit by intentionally assigning stretch tasks and providing learning support
Risk Owners	Team Lead / Project Manager
Probability	Low
Impact	Very Low
Status	Skill development tasks being planned into

	the schedule
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Number	R5
Rank	4
Risk	Serverless Migration
Description	Shift to serverless architecture may improve scalability and cost efficiency
Category	Technical
Root Cause	Availability of modern AWS services supporting serverless models
Triggers	Exploration of AWS Lambda, DynamoDB, etc. spikes
Potential Responses	Share opportunity by involving DevOps and cloud partners early
Risk Owners	Solution Architect
Probability	Moderate
Impact	High
Status	Being reviewed in technical roadmap meeting

Number	R6
Rank	4
Risk	Scoping Tools Development
Description	Development of reusable tools or templates to assist in future scoping activities
Category	Opportunity
Root Cause	Poor hosting configuration or lack of redundancy
Triggers	Repeated scope issues, process documentation efforts
Potential Responses	Enhance by allocating time for tool development during low-pressure phases
Risk Owners	Business Analyst
Probability	Moderate
Impact	Moderate
Status	Early-stage exploration underway

8.3 Qualitative Risk Analysis

IMPACT SCALE TABLE

Risk	Relative/ Numerical Scale				
	Very Low/0.05	Low/0.1	Moderate/0.2	High/0.4	Very High/0.7
R1					
R2					
R3					
R4					
R5					
R6					

PROBABILITY SCALE TABLE

Likelihood		Description
Relative	Numerical	
Very Low	0.1	Highly unlikely to occur
Low	0.3	Will most likely not to occur
Moderate	0.5	Possible to occur
High	0.7	Likely to occur
Very High	0.9	Highly likely to occur

PROBABILITY IMPACT MATRIX

Probability	Risk (Impact x Probability)				
Very High / 0.9				R1	
High /0.7			R2, R3		
Moderate / 0.5			R6	R5	
Low / 0.3	R4				
Very Low / 0.1					
Impact	Very low/0.05	Low/0.1	Moderate/0.2	High/0.4	Very High/0.7
R1	$= 0.9 \times 0.4$ $= 0.36$				
R2	$= 0.7 \times 0.2$ $= 0.14$				
R3	$= 0.7 \times 0.2$ $= 0.14$				
R4	$= 0.3 \times 0.05$ $= 0.015$				
R5	$= 0.3 \times 0.4$ $= 0.12$				
R6	$= 0.5 \times 0.2$ $= 0.1$				

8.4 Risk Response Plan

No .	Risk	Response Strategy	Planned Actions	Justification
1	Scope Creep	Avoidance	Enforce strict change control and scope boundaries with formal approval	Can be actively avoided with a structured scope management plan.
2	Google Maps API Overuse	Mitigation	Engage early with API providers, allocate buffer time, implement mock API testing	Integration is a key dependency and cannot be avoided or transferred. Early mitigation reduces failure chances.
3	AWS Hosting Overages: Downtime Mitigation	Transference & Mitigation	Use high-availability cloud services (transference), implement real-time monitoring and auto-scaling (mitigation)	Leverage external SLA-backed services, while retaining internal resilience.
4	Team Skill Growth	Exploitation	Assign learning tracks aligned with project needs, encourage cross-training	Skill growth is positive, but if misaligned, it could lead to inefficiencies. Managed proactively, it boosts capability.
5	Serverless Migration	Exploitation	Pilot migration on non-critical modules, measure cost & performance	Serverless can reduce cost and downtime if adopted strategically. Turning this into an advantage requires planning.
6	Scoping Tools Development	Exploitation	Train the team to use tools for clearer requirements gathering	Tool adoption boosts project clarity but also needs a learning curve to avoid initial misuse or delay.

9.0 Procurement Management Plan

The procurement management plan outlines the process and decisions regarding the acquisition of external tools and services required for the successful delivery of the Automated Order Routing System. As this system includes integration with various third-party APIs, the use of cloud infrastructure, and licensed development tools, careful consideration has been taken to select reliable and cost-effective solutions that align with the project objectives and budget constraints.

9.1 Procurement Item

Microsoft Project Plan 3 will be acquired as the project management software. It offers powerful scheduling and task tracking capabilities, supporting the Agile methodology adopted for the system's development. The tool enables better visibility and control over project timelines, resources, and milestones.

Google Maps API will be licensed to provide real-time location tracking and routing functionality. By using Google Maps' infrastructure, the development team can leverage precise GPS data, traffic updates, and mapping capabilities without building a custom navigation solution. The API supports functions critical to the system, such as driver location updates and dynamic route optimization.

AWS EC2 (Elastic Compute Cloud) will be used to host backend services and machine learning models. The t3. medium instance offers a cost-effective yet scalable infrastructure, ideal for supporting web-based applications under varying traffic loads. AWS's global reach and high availability ensure system performance and reliability.

Table:

No.	Item	Quantity	Unit Price (RM)	Subtotal (RM)	Vendor
1	Microsoft Project Plan 3	1	127.57	127.57	Microsoft Corp.
2	AWS EC2 t3.medium	1	201.60 (per mo.)	1,209.60	Amazon Web Services
3	Google Maps API	1	0.00 (Free tier)	0.00	Google

Total: RM 1,337.17

9.2 Supplier selection evaluation

Table 1: Proposal Evaluation Matrix

Criteria	Weightage (%)	Describe
Technical merit (fit for purpose)	60%	Solution must support real-time analytics, ML hosting, and API integrations.
Capability of supplier to deliver	25%	Vendor must guarantee uptime (SLA $\geq 99.9\%$), support, and compliance.
Value for money	15%	Cost must align with the budget (RM300k cap) and offer transparent pricing.
TOTAL	100%	-

Item: AWS EC2 Hosting Platform

Criteria / Weight	AWS (Amazon Web Services)		Azure (Microsoft)		Exabytes (Malaysia)	
	Rating	Score	Rating	Score	Rating	Score
Technical Merit (60%)	9 Reason: Highly scalable & secure platform	54	7 Reason: Competitive cloud services	42	6 Reason: Limited scalability	36
Capability to Deliver (25%)	9 Reason: Proven global infrastructure	22.5	8 Reason: Strong enterprise integration	20	6 Reason: Local-focused	15
Value for Money (15%)	8 Reason: Reasonable RM/hourly rate	12	6 Reason: Higher enterprise pricing	9	9 Reason: Lower cost, but fewer features	13.5
Total Score (100%)	88.5		71		64.5	

Conclusion:

AWS is selected due to its superior scalability, global reliability, and competitive pricing. It aligns best with the project's need for high availability and real-time processing.

Item: Microsoft Project Plan

Criteria / Weight	Microsoft Project		Trello Premium		ClickUp Pro	
Technical Merit (60%)	9 Reason: Rich PM tools for Agile	54	6 Reason: Visual but lacks Gantt	36	7 Reason: Flexible but less formal	42
Capability (25%)	9 Reason: Microsoft support, cloud	22.5	7 Reason: Decent user base	17.5	8 Reason: Popular SaaS PM tool	20
Value for Money (15%)	8 Reason: Full-featured, affordable	12	9 Reason: Low monthly fee	13.5	7 Reason: Costlier at scale	10.5
Total Score (100%)	88.5		67		72.5	

Conclusion:

Microsoft Project is selected for its comprehensive Agile support, integration with other tools, and cost-effectiveness for project management needs.

Item: Google Maps API

Criteria / Weight	Microsoft Project		Trello Premium		ClickUp Pro	
Technical Merit (60%)	9 Reason: Rich PM tools for Agile	54	6 Reason: Visual but lacks Gantt	36	7 Reason: Flexible but less formal	42
Capability (25%)	9 Reason: Microsoft support, cloud	22.5	7 Reason: Decent user base	17.5	8 Reason: Popular SaaS PM tool	20

Value for Money (15%)	8 Reason: Full-featured, affordable	12	9 Reason: Low monthly fee	13.5	7 Reason: Costlier at scale	10.5
Total Score	88.5		67		72.5	

Conclusion:

Google Maps API is selected for its unmatched accuracy, real-time traffic updates, and free-tier availability, which reduces initial costs.

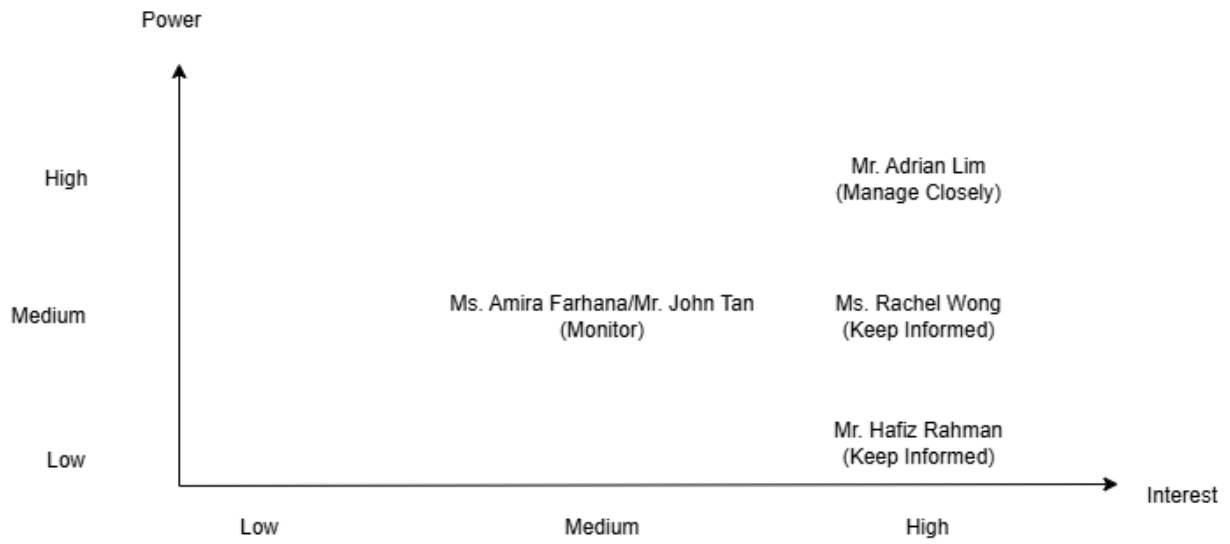
10.0 Stakeholder registry

Stakeholder management is essential to ensure that all parties involved in the project are aligned with its goals and deliverables. This section outlines the key stakeholders, their roles, levels of influence and interest, and strategies to ensure effective communication and engagement throughout the project lifecycle. Clear stakeholder identification and proper management help minimize resistance, manage expectations, and enhance project success.

10.1 Stakeholder Registry and Management Strategy

Name	Position	Internal/ External	Project Role	Contact Info	Interest	Influence	Management Strategy
Mr. Adrian Lim	CTO of Company XX	Internal (Client)	Project Sponsor	adrian.lim@xxlogistics.com	High	High	Engage closely, frequent progress updates, and engage in key decision-making
Ms. Rachel Wong	Project Manager	Internal	Project Coordinator	rachel.wong@xxlogistics.com	High	Medium	Frequent updates and sprint reviews
Mr. John Tan	AWS Account Manager	External (Supplier)	AWS Cloud Vendor Representative	john.tan@amazonaws.com	Medium	Medium	Regular check-ins and budget monitoring
Mr. Hafiz Rahman	Driver Representative	Internal (Client)	End-User Representative	hafiz.rahman@xxlogistics.com	High	Low	Surveys, pilot training feedback loops
Ms. Amira Farhana	QA Manager	Internal	Quality assurance, testing	amira.f@xxlogistics.com	Medium	Medium	Weekly testing syncs, raises critical issues

10.1.1 Power / Interest Grid:



11.0 Conclusion

In today's fast-paced logistics world, efficiency, precision, and responsiveness are essential. Company XX's reliance on a manual driver assignment system has significantly hindered its ability to expand, serve clients reliably, and compete with more technologically advanced logistics companies. Manual coordination's limitations—such as increased human error, a lack of real-time visibility, and variable delivery performance—are no longer viable in a business that wants to develop and form long-term partnerships with significant e-commerce giants like Shopee.

This proposal suggests a transformative step forward by implementing an Automated Order Routing System. By combining modern data analytics and machine learning, the system will intelligently assign orders to the best drivers or fulfilment centers based on real-time demand trends, driver location, warehouse capacity, and delivery performance. This data-driven strategy improves operational efficiency while simultaneously increasing transparency, accountability, and customer satisfaction.

Using agile methodology, the project was broken into sprints that included core components such as machine learning, and UI/UX development. There was a complete Work Breakdown Structure (WBS) and Gantt chart that was followed in the project timeline and resource allocation to make it open and trackable.

Cybersecurity will also be a key priority to guarantee that sensitive customer information and location data are managed in strict accordance with data protection regulations. This is especially important when growing operations to accommodate large corporate clients who expect a high level of trust, dependability, and regulatory compliance.

For the procurement planning, Microsoft Project, AWS, and Google Maps API were selected as the external parties in this project.

The entire system would be created, tested, and deployed over a six-month project period, with a reasonable yet ambitious budget of RM 274680.10 including 10% management reserves with RM 249709.17 cost baseline. This ensures that the project is both technologically sound and financially sustainable.

To summaries, the proposed Automated Order Routing System is more than just a technical enhancement; it is a strategic investment in Company XX's future. It enables the organization to shift from reactive, manual procedures to a proactive, intelligent, and scalable logistics framework. Once implemented, the system will drastically cut inefficiencies, improve delivery speed and accuracy, and provide opportunities for lucrative partnerships with large-scale clients. By adopting innovation immediately, Company XX establishes itself as a competitive and modern logistics provider capable of meeting the changing demands of the digital economy.

12.0 Self-Evaluation

Tan Sze Yee: From this project and this module, I have gained a detailed understanding of planning a software project, and completing the assignment helped me to implement project management theories in practice. From all the concepts that I had learned throughout this module, one of the most significant things that I learned is the Work Breakdown Structure (WBS) and cost management, specifically in establishing the cost baseline and knowing clearly how contingency and management reserves are established. This helps me to manage future software projects.

Retaj Ahmed: During this project, we had regular group meetings to follow up on project progress and share insights which enhanced my communication skills. I learned how to start working early to avoid any delays, and to break down large tasks into smaller ones to finish my associated tasks on time. I learned how to work comprehensively and always follow up on project progress to have an inclusive understanding and always be on the same page with my teammates and align with project goals. I am grateful that my teammates were responsible and helpful throughout the whole assignment.

Adwin Chee Hansen: This project enhanced my knowledge on the Microsoft Project software. Initially, I had issues on the structure as well as the different fields of the application. The project required semi-advanced knowledge which required me to learn the different fields and functions of the application. The project also helped me manage my time better, working on the GANTT chart, allowed me to produce a timeline that is the most efficient while maintaining a balanced workload. Overall, the project went smoothly, and I was able to complete my task easily, mostly because Retaj was able to produce a really good WBS.

Aloysius Ratnam: Throughout this project, I gained a deeper understanding of how important proper documentation and task delegation are in ensuring smooth project execution. I learned how to stay organized by tracking risk items and updating them as the project progressed. I also became more comfortable using project management tools and working under deadlines, which has strengthened my ability to contribute effectively in a team setting. My main task was to contribute to the risk management plan. This taught me how to identify potential project hazards, assess their impact and probability, and develop appropriate response measures. This work increased my awareness of proactive planning and the significance of anticipating obstacles throughout project execution.

Brendon Chong: This project helped me apply project management concepts like Work Breakdown Structure (WBS) and cost estimation in practice. Through teamwork, I improved my communication and time management by breaking tasks into smaller steps and tracking progress. Using tools like Microsoft Project and GANTT charts enhanced my technical skills in scheduling. Contributing to risk management also taught me to identify and mitigate potential issues early. Overall, the experience strengthened my ability to plan, collaborate, and execute projects effectively.

Solomon Tan Karl-El: Throughout this project, I grew in my appreciation for project managers as I can see the unending responsibilities that they have in every knowledge area. Personally, I have

never had any experience with project management, so even things like coming up with a communication plan and writing the 2.0 project scope statement was a completely new experience for me. The learning curve was steep, and there were many times I felt like giving up understanding and learning how it works, but I told myself that I am not doing this just to complete the assignment, but to become a potential project manager myself in the future. So overall, it was a meaningful experience, and I am positive that the skills I have learnt here will translate into the working world in the future.

Cheong Kai Liang: This assignment project has deepened my concept of how project manager playing the main role in a project, in the stage of planning, coordination, and decision-making. By my contributed part (5.0 &6.0), I learned how quality tools and human resource planning directly impact the success of a system like the one we doing (AORS). Through the research on the project has showed me the importance of aligning roles, timelines and performance goals realistically. This experience gave me real insight into how IT projects are managed in the industry and how crucial clear structure and planning are to a successful software outcome that meet business requirement and stakeholder behavior.

13.0 Task Distribution Table

Name	Task
[REDACTED]	4.0 Project Cost Management
RETAJ AHMED [REDACTED] [REDACTED]	2.2 Project Implementation Approach 2.3 Work Breakdown Structure 2.4 Key Project Milestone
[REDACTED]	3.0 Project Time Management 3.1 GANTT Chart
[REDACTED]	8.0 Project Risk Management Plan 11.0 Conclusion
[REDACTED]	1.0 Introduction 2.1 Project Description 7.0 Project Communication Plan
[REDACTED]	5.0 Project Quality Management Plan 6.0 Project Human Resources Management Plan
[REDACTED]	9.0 Procurement Management Plan 10.0 Stakeholder Register

14.0 References

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15.0 Others

15.1 Appendix

PMP_Group2_Turnitin

ORIGINALITY REPORT

5%

SIMILARITY INDEX

2%

INTERNET SOURCES

0%

PUBLICATIONS

4%

STUDENT PAPERS

PRIMARY SOURCES

1

Submitted to Taylor's Education Group

Student Paper

2%

2

Submitted to INTI Universal Holdings SDM
BHD

Student Paper

1%

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Submitted to Swinburne University of
Technology

Student Paper

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Student Paper

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Assessment Criteria

Assignment contributes **30%** of the overall marks. Your efforts will be assessed based on the following criteria:

Project management plan elements						
Scope statement (project charter)	25%	All requirements met, appropriately used, excellent representation of element.	All requirements met, very appropriately used. Little or no attempt was made to conform to requirement. Does not meet the minimum required elements.	Most requirements met, appropriately used, formatting could be improved further. Some requirements met, missing vital details.	Little or no attempt was made to conform to requirement. Does not meet the minimum required elements.	
Work breakdown structure	10%					
GANTT chart	10%					
Project cost and budget (including contingency and management reserve)	20%					
Quality objective and tool	5%					
Project team chart	5%					
Communication plan	5%					
Risk identification and response plan	10%					
Procurement plan	5%					

Stakeholder registry and management strategy	5%						
TOTAL (100%)							
WEIGHTED TOTAL (30%)							